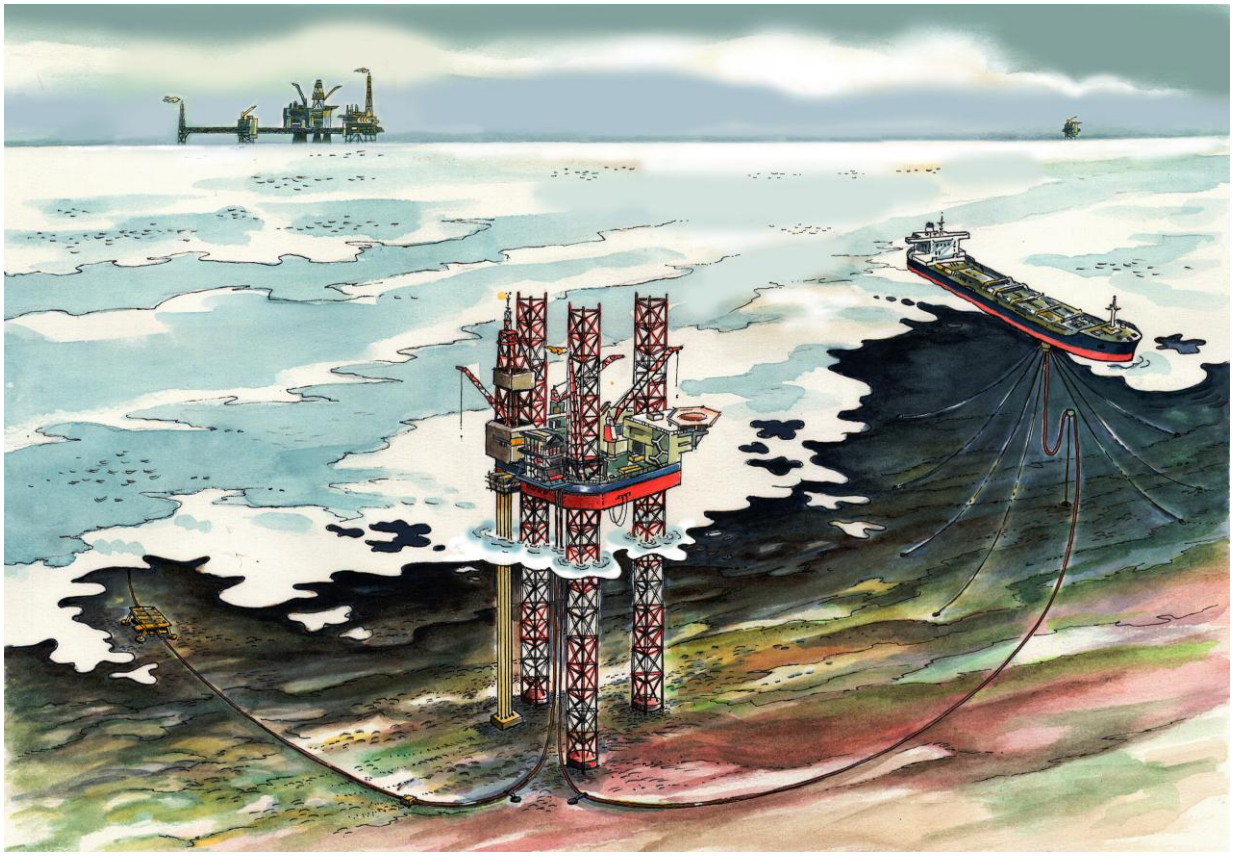

FINAL WELL REPORT

Completion

License: NO PL046BS

Well: NO 15/9-F-11 B

Statoil

Title:		
FINAL WELL REPORT Completion License: NO PL046BS Well: NO 15/9-F-11 B		
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1 General well data

Objective:
The completion is to provide for production of oil from the Hugin reservoir.
The completion is designed for easy access when it comes to well interventions and later P&A operations. The well interventions can be PLT and RST runs, plug setting or straddle installation to shut off water production and re-perforation operations, However interventions are not limited to mentioned operations.
The well has been brought on production after being perforated using tractor conveyed orientated guns.
Planned well life is 10 years.

Well Data:

AFE # (optional)	
Licence number:	NO PL046BS
Well name:	NO 15/9-F-11 B
Slot:	F11
Type of well:	Oil producer
Water depth / airgap:	91,0 m MSL / 54.9 m RT
Distance RT - LRR: (optional)	-
Wellhead / PGB: (optional)	-
Primary objective:	Oil production from Hugin formation
Completion type:	7” monobore

Coordinates at Wellhead Level (as installed 27.06.07)				
	Structure Centre		Slot Centre F-11B	
	N	E	N	E
UTM:	6 478 563.52	435 050.02	6 478 568,18	435 049,10
Geographic	58° 26' 29.807"	1° 53' 14.929"	58° 26' 29.957"	1° 53' 14.867"
Rectangular (from structure centre)	0	0	4,66	-0,92

Completion Strategy:
The F-11B IOR completion is a 7” monobore design with 7” production tubing and 7” cemented liner.

The completion incorporates: 7” x 9 5/8” production packer, single DHG for reservoir pressure and temperature monitoring and a TRSCSSV down hole safety valve.
The 7” liner is placed in the high permeable Hugin sand-reservoir in the main Volve field.
The well has been perforated using wirelinetractor conveyed oriented guns.

The wells are tied into the platform with 14” Production Risers from subsea WH to surface WH.
Prior to the completion a 10 ¾” surface tieback casing is installed from the subsea WH to the surface WH inside the 14” Production Riser, connecting the subsea and surface WH. The entire riser system is set in app. 150 ton tension provided by the hydraulic tension system.

Perforation:
The well has been perforated using oriented guns on wire-line tractor. Estimated perforated interval in 7" liner is 8 runs, approximately 12,5 meter pr. run.
The perforation guns were 4 ½" OrientXact, PowerPlus Low debris Charges, phasing 10, with 4 shots per foot.

Operational Sections:

Section	Start time	End time	Rig name
NO 15/9-F-11 B Wellbore preparation	16.Jun.2013 13:30	21.Jun.2013 07:00	MÆRSK INSPIRER
NO 15/9-F-11 B Other	21.Jun.2013 07:00	21.Jun.2013 21:00	MÆRSK INSPIRER
NO 15/9-F-11 B Other	21.Jun.2013 21:00	28.Jun.2013 01:00	MÆRSK INSPIRER
NO 15/9-F-11 B Tie-back	28.Jun.2013 01:00	30.Jun.2013 07:15	MÆRSK INSPIRER
NO 15/9-F-11 B Other	30.Jun.2013 07:15	30.Jun.2013 14:15	MÆRSK INSPIRER
NO 15/9-F-11 B Run Completion	30.Jun.2013 14:15	04.Jul.2013 20:00	MÆRSK INSPIRER
NO 15/9-F-11 B Install Tubing Hanger	04.Jul.2013 20:00	06.Jul.2013 12:30	MÆRSK INSPIRER
NO 15/9-F-11 B Install X-mas Tree	06.Jul.2013 12:30	09.Jul.2013 00:45	MÆRSK INSPIRER
NO 15/9-F-11 B Perforation	09.Jul.2013 00:45	21.Jul.2013 23:00	MÆRSK INSPIRER

2 Dispensations

Well is completed without any dispensation.

3 Health, environment, safety and quality (HES&Q)

3.1 RUH

Type (colour code)	Synergi code	Number of
Yellow	3	1
Sum		1

3.1.1 RUH details

WELLBORE: NO 15/9-F-11 B

Synergi no	Hazard	Description
1364582	3	Synergi 1364582 (Closed) - 14.Jun.2013 - Non-conformities - Tong marks on 7" 29# SM Cr13 110 Vam TOP HT pin ends During running of the 7" liner on well 15/9-F-11 B a number of joints had to be laid out. After visual inspection damages similar to tong marks from a bucking unit were identified on some of the joints. It is suspected that these joints have been damaged onshore.

3.2 Experience listing

Subject:	Claxton tool not able to release		
Section:	NO 15/9-F-11 B, Other		
Rep date:	27.Jun.2013		
Keywords:	WELLHEAD/ASA		
Downtime:		**Pot time improvements:**	
Comp inv:	GE Vetco		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

Description:
Claxton tool not able to release due to low pressure (1500 psi) applied on the piston. Increased pressure to 1800 psi and released tool without any trouble.
Immediate solution:
Incresed pressure from 1500 psi to 1800 psi
Future recommended solution:
Proper procedures

Subject:	F-11 Tie Back installation		
Section:	NO 15/9-F-11 B, Tie-back		
Rep date:	29.Jun.2013		
Keywords:	WELLHEAD/ASA		
Downtime:		**Pot time improvements:**	
Comp inv:	Vetco		
References:			
Synergi no:		**Cost:**	
Synergi desc:			

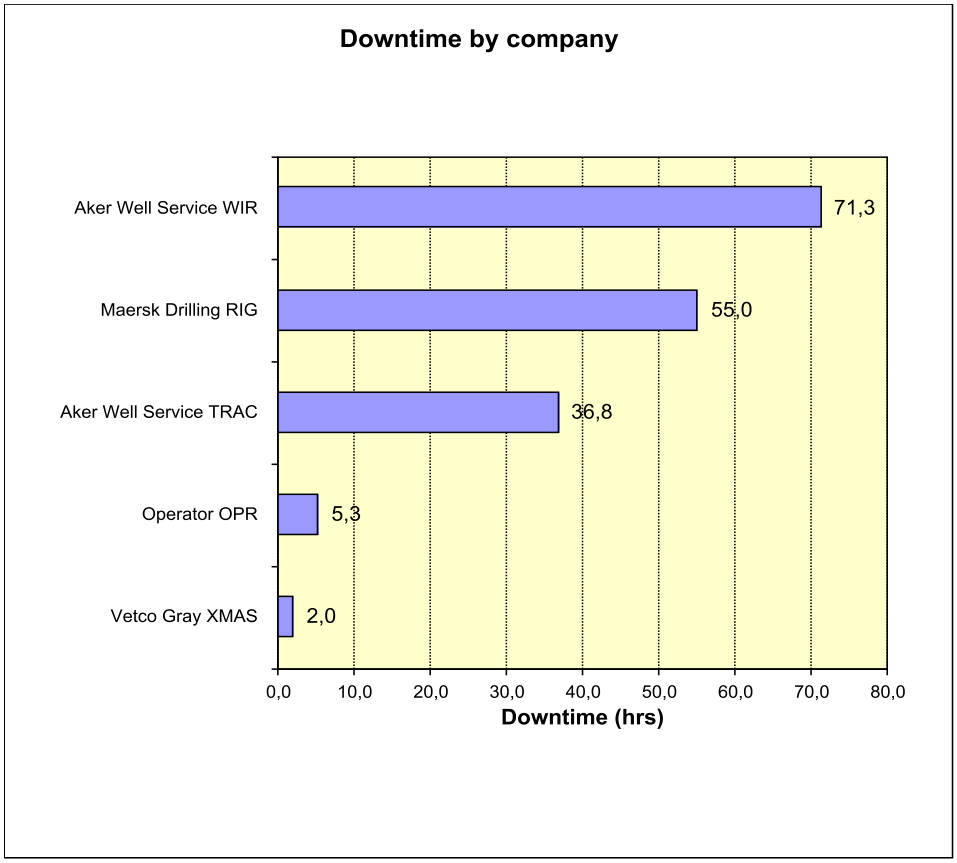
Description:
Operation:
When running in with the tie back assembly, we lowered it slowly, recorded down and up weights, then tagged the thread of the subsea casing hanger and set down 2 ton of landing string weight and measured from top of the joint in the rotary to the rotary table, we then marked the pipe with a vertical line and rotated the string using the TDS and elevators 1 full turn. We then measured again and compared figures which indicated we had entered the thread, we then proceeded to rotate another full turn and re-measure to confirm and the indication was positive that we were entering the threads. We then removed the elevators and continued to rotate the casing with a chain tong to remove the casing hanger RT, however, once a few rotations were performed (with difficulty) we put the elevators back onto the string and picked up 1 ton and rotated out the casing hanger running tool.

Recommendations for consideration:
To rotate tie back as stated above but after proof by measurement that we were in the threads we would continue with rotations using the TDS and elevators in a controlled manner, this would eventually build a little torque until a point was reached where the casing hanger RT broke out and we would recover to surface in one smooth operation. We would then continue to follow the DOP as planned to finish the setting and testing plan of the operation.

Immediate solution:

Future recommended solution:

3.3 Incidents by service and company



SERVICE COMPANY: Operator OPR Operator
WELLBORE: NO 15/9-F-11 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
24.jun.2013	RIG-E20 Riser system					5,3	100	5,3
			Total	0				5,3

SERVICE COMPANY: Rig Operations RIG Maersk Drilling
WELLBORE: NO 15/9-F-11 B

Incident start	Failure code	Synergi	Title	Quality cost	Downtime

time		no		NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
16.jun.2013	RIG-E07 Top drive	1365331	Broke release failure on top drive	2989175	1568092	45,8	100	45,8
20.jun.2013	RIG-E341 Vert pipe handling system	1365335	Top drive collided with stand in the pipe racking machine and bent top joint.	824600	0	0,0	100	0,0
27.jun.2013	RIG-E03 BOP stack/valves	1366412	Attempted to pressure test annular BOP on stump.	3000	8471	8,5	100	8,5
01.jul.2013	RIG-E342 Drillfloor tub handl system	1366882	HTS belt sensor defect.	41206	1977888	0,5	100	0,5
02.jul.2013	RIG-E345 Elevator	1366883	Observed hydraulic leak from elevator fitting, replaced same	20500	1968000	0,3	100	0,3
			Total	3878481				55,0

SERVICE COMPANY: Surf. WH/X-mas tree XMAS Vetco Gray
WELLBORE: NO 15/9-F-11 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
29.jun.2013	XMAS-E01 Hanger pack off	1367522	Not able to lock RT into riser due to internal hydraulic leak i RT.	164812	1977744	2,0	100	2,0
			Total	164812				2,0

SERVICE COMPANY: Tractor TRAC Aker Well Service
WELLBORE: NO 15/9-F-11 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
10.jul.2013	TRAC-E03 Electronic tractor failure	1368443	Wireline Tractor Failure	3000	3692	19,5	100	19,5
13.jul.2013	TRAC-E01 Cable head	1368445	Lost Communication with Perforation assembly	3000	4174	17,3	100	17,3
			Total	6000				36,8

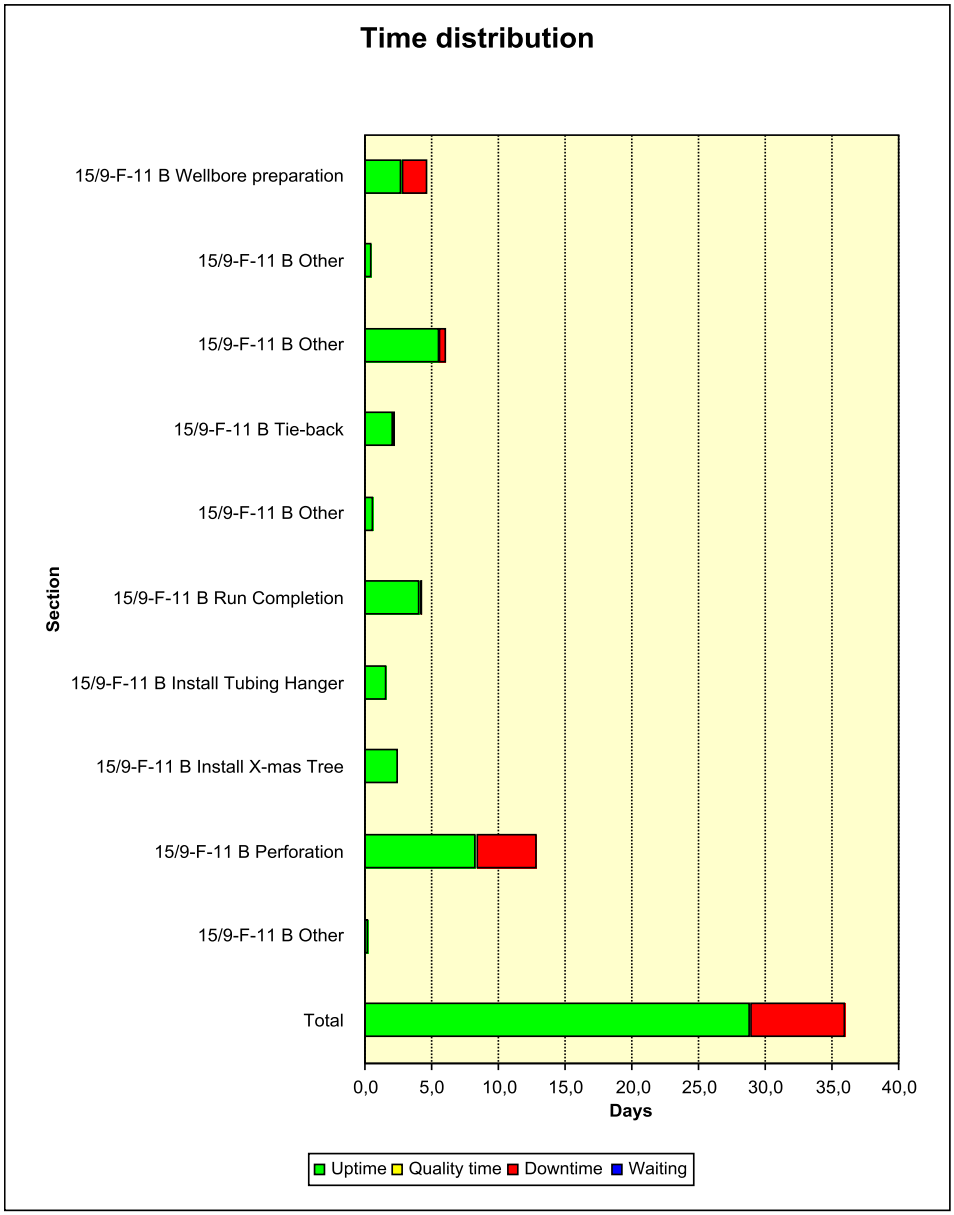
SERVICE COMPANY: Wireline WIR Aker Well Service

WELLBORE: NO 15/9-F-11 B

Incident start time	Failure code	Synergi no	Title	Quality cost		Downtime		
				NOK	NOK/d	Total hrs	Comp share %	Comp share hrs
11.jul.2013	WIR-E10 Winch/Power pack					0,8	100	0,8
11.jul.2013	WIR-E02 Weakpoint	1368191	Wire Line Cable Disconnected from Tool String.	453530	2721180	4,0	100	4,0
15.jul.2013	WIR-E02 Weakpoint	1368632	Lost Communication to Perforation BHA	3000	1083	66,5	100	66,5
			Total	456530				71,3

3.4 Time distribution

		Length	Budget		TL		Actual		Ops
Section	Start time	m	hrs	days	hrs	days	hrs	days	(f)
NO 15/9-F-11 B Wellbore preparation	16.Jun.2013 13:30	90,0	99,0	4,1	59,0	2,5	113,5	4,7	59,7
NO 15/9-F-11 B Other	21.Jun.2013 07:00	90,0	19,0	0,8	10,0	0,4	14,0	0,6	100,0
NO 15/9-F-11 B Other	21.Jun.2013 21:00	90,0	216,0	9,0	179,5	7,5	148,0	6,2	90,7
NO 15/9-F-11 B Tie-back	28.Jun.2013 01:00	90,0	55,5	2,3	29,0	1,2	54,3	2,3	96,3
NO 15/9-F-11 B Other	29.Jun.2013 23:00	90,0	15,0	0,6	8,0	0,3	15,3	0,6	100,0
NO 15/9-F-11 B Run Completion	30.Jun.2013 14:15	90,0	100,0	4,2	96,0	4,0	101,8	4,2	99,3
NO 15/9-F-11 B Install Tubing Hanger	04.Jul.2013 20:00	90,0	87,0	3,6	57,0	2,4	40,5	1,7	100,0
NO 15/9-F-11 B Install X-mas Tree	06.Jul.2013 12:30	90,0	94,0	3,9	87,0	3,6	60,3	2,5	100,0
NO 15/9-F-11 B Perforation	09.Jul.2013 00:45	90,0	206,0	8,6	226,0	9,4	310,3	12,9	65,2
Sum			891,5	37,1	751,5	31,3	858,0	35,7	



The graph above is based on the following details:

Section	Downtime	Uptime	Quality time	Waiting time	Total time
15/9-F-11 B Wellbore preparation	1,9	2,8	0,0	0,0	4,7
15/9-F-11 B Other	0,0	0,6	0,0	0,0	0,6
15/9-F-11 B Other	0,6	5,6	0,0	0,0	6,2
15/9-F-11 B Tie-back	0,1	2,2	0,0	0,0	2,3
15/9-F-11 B Other	0,0	0,6	0,0	0,0	0,6
15/9-F-11 B Run Completion	0,0	4,2	0,0	0,0	4,2

Section	Downtime	Uptime	Quality time	Waiting time	Total time
15/9-F-11 B Install Tubing Hanger	0,0	1,7	0,0	0,0	1,7
15/9-F-11 B Install X-mas Tree	0,0	2,5	0,0	0,0	2,5
15/9-F-11 B Perforation	4,5	8,4	0,0	0,0	12,9
Total	7,1	28,6	0,0	0,0	35,7

4 Activity highlights

4.1 15/9 F-11B IOR, Liner (12.06.2013 20:15)

START: 12.Jun.2013 20:15 - 4770 mMD
END : 16.Jun.2013 13:30 - 4770 mMD

OBJECTIVE
-Run 7" liner, and cement over it's entire length.

SUMMARY

The 7" liner was run on 5 1/2" DP. The average speed was 285 m/hr in cased hole and 230 m/hr in Open hole. From 3435 mMD established circulation with 400 lpm and 32 bar and wash down to 4769 mMD. Set liner hanger 100 meter above the 9 5/8" casing shoe according to procedure. Circulated 2 open hole volume with 1200 LPM and 86 bar. 11.2 m3 of 1,92 cement slurry was pumped according to procedure with no losses. Set liner hanger packer according to procedure. Tested the well to 345 bar successfully.

4.2 15/9 F-11B IOR, Well bore preparation (16.06.2013 13:30)

START: 16.Jun.2013 13:30 - 4770 mMD
END : 21.Jun.2013 07:00 - 4770 mMD

OBJECTIVE
- Clean BOP, riser and subsea wellhead in mud.
- Dress 7" liner PBR.
- Drift and clean 10 ¾" x 9 ⅝" casing and 7" liner.
- Inflow test well with Well Commissioner
- Displace well to 1,03sg NaCl packer fluid.
- Leak test well to 345 bar/1.03 sg NaCl packer fluid

SUMMARY

-RIH w/ cleanout and drift assemblies. Scrape the packer setting depth from 3017 to 3354 mMD. Polished the 7" PBR. Conditioned the mud and Inflow test the liner with 93 bar. Pumped wash train and displace the well to 1.03 completion fluid with 1700 lpm / 160 bar. Pressure tested the well to 345 bar and pulled the clean-up string out of hole.

-RIH w/jet sub cleanout assy. Washed WH, Riser and BOP reciprocating the washing assy. across the specified areas, w/ 1500 lpm. Riser was displaced by sea water. Clean up assembly pulled out of hole.

4.3 15/9 F-11B IOR, Temporary abandonment (SAFELOCK plug) (21.06.2013 07:00)

START: 21.Jun.2013 07:00 4770 mMD
END : 21.Jun.2013 21:00 4770 mMD

OBJECTIVE
- Install 9 5/8” SAFELOCK plug in first joint below 9 5/8’’ x 10 ¾’’ x-over, at +/- 620 m MD.
- Leak test SAFELOCK plug to 345 bar from below.
- Leak test SAFELOCK plug to 345 bar from above.
- Retrieve 10-3/4” Wearbushing from Subsea WH
- Install Lock Down Sub in Subsea WH

SUMMARY

Picked up and made up Mechanical plug (SAFELOCK). Ran in hole with SAFELOCK plug on 5 1/2" heavy weight drill pipe down to 607 mMD. Established circulation with 1100 lpm and washed setting area. Set the plug at first joint of 9 5/8¿ (605 mMD) according the ARCHER procedure. Setting was verified by applying 10 MT overpull. Plug pressure tested 345 bar from bellow and above. Spotted 5m3 high visc pill on top of SAFELOCK plug. Displaced riser and BOP to seawater with 2200 lpm and 10 bar. Pulled out of hole with running tool. Picked up and ran in hole with wearbushing retrieval tool and jet sub on 5 1/2" heavy weight drill pipe. Jetted wellhead area and retrieve the wearbushing according GE (VETCO) procedure. Pulled out of hole with wearbushing. Pick up and ran in hole with 9.5" drift on 5 1/2" drill pipe to 575m MD. Pulled out of hole the same. Picked up and ran in hole with lockdown sub. Installed the Lock down assembly according to GE procedure. Lock down assembly tested to 207 bar. Pulled out of hole with the running tool assembly.

4.4 NO 15/9-F-11 B, Other (Retrieve HPDR and install PR and surface WH) (21.06.2013 21:00)

START: 21.Jun.2013 21:00 4770 mMD
END : 28.Jun.2013 01:00 4770 mMD

OBJECTIVE
- Nipple down BOP and Diverter assembly
- Pull and lay out HPDR
- Run and installation of Production Riser

SUMMARY

Pulled master bushings and outer ring and installed lift eyes to diverter. Pulled diverter and landed on set back. Pulled slick joint and landed in set back. Installed master bushings and ring. Nippled down BOP. Rigged down tension cylinders and tension ring. Rigged up sheaves for running guide wires. Ran hot stab to seabed and installed same in tieback connector with ROV. Pressured up hot stab connection to 3000 psi. Unlatched tieback connector and picked riser off wellhead. Retrieved hotstab and reinstalled dummy stab. Pulled up with tieback connector and high pressure drilling riser. Racked back 5 1/2" drill pipe stand. Installed riser spider and function tested same. Picked up BOP joint and closed spider dogs. Installed C-plate and side door elevator onto spider. Landed off BOP joint in side door elevator on top of spider. Laid out BOP joint and claxton tool. Pulled High pressure Drilling Riser in double joint. Retrieved guidelines from seabed. Pulled lower riser pup joint.

93.5 hours waited on weather for running production riser. Picked up tieback connector and stress joint and landed same in rotary.
Picked up and made up riser pup joint to stress joint.
Installed fairings.
Lowered production riser through rotary to test connector. Function tested tieback connector. Picked up standard riser joint #1 made up to riser pup joint. Pressure tested flange connection. Installed flange protection fairings. Installed guide wires and ROV hooks. 7.75 hours waited on weather for running production riser. Picked up and ran the production riser in standard joints. Each flange connection pressure tested to 3000 psi through the test port. Lowered riser and sat in riser spider. Disconnected riser running tool from riser tension joint and cleaned seal face. Made up pre-made riser running tool to tension joint. Made up pre-made riser running tool to tension joint. Lowered riser to safe distance above riser landing guide funnel. Verified VGX ring installed with ROV. Stabbed riser into vertical guide and onto WH. Removed riser guide funnel. Landed off tieback connector on wellhead with 15 Ton. Locked connector by applying 3000 psi until stroke indicator indicated fully set. Performed 10 Ton overpull test to verify connector fully latched. Pressure tested production riser to 200 bar. Cut and bewelled production riser. Ran and installed the multi bowl and BOP joint assembly. Nippled up the BOP.

4.5 NO 15/9 -F-11B IOR, Install 10 ¾” tie 11 B, Tie-back (28.06.2013 01:00)

START: 28.Jun.2013 01:00 4770 mMD
END : 30.Jun.2013 07:15 4770 mMD

OBJECTIVE
- Install Bowl Protector in surface wellhead.
- Clean the riser and BOP
- Pull Annulus insert
- Retrieve bowl protector from surface wellhead.
- Run 10 ¾”3/4" surface tie-back casing on spear and sting into sub sea WH with tie-back adaptor
- Run the adjustment sub torque tool and apply tension to tie-back string
- Leak test 10 ¾” tie-back casing to 345 bar against SAFE LOCK plug using Vetco casing hanger running tool.
- P/U and RIH surface bowl protector.

SUMMARY

Installed bowl protector. RIH with jetting sub and clean subsea wellhead and displace sea water with 3000 liter/min, 25 bar, 5 rpm. Pulled out of hole with jetting sub. RIH with annulus insert retrieval tool. Retrieved annulus insert at 139,95 meter with 6000 Nm. Displaced riser to 1.03sg inhibited brine. POOH with annulus insert retrieval tool and insert. Rigged up 10 3/4" handling equipment for tie-back casing. Picked up 10 3/4" tie-back connector and made up the same to first 10 3/4" casing joint. RIH with 10 3/4" tie-back string as per running tally from surface to 139,88 meter. Tagged land off point with 3 Ton. Engaged the Tie-back connector according to Drill-quip procedure. Rigged down handling equipment for 10 3/4" tie-back casing. Picked up casing hanger landing string and RIH to top of casing hanger at 20,5 meter. Rig up the swedge and pressure tested tie-back adapter to 345 bar. Picked up adjustable sub setting tool and RIH with drill pipe to 28,3 meter . After two mis-runs and repositioning the adjustable sub set by 9000 Nm torque. Pressure tested adjustable sub to 345 bar. Picked up 10 3/4 MS-T seal assembly running tool , ran in hole with MS-T seal to 22,75 m. Set MS-T seal according to the procedure. Tested MS-T seal assembly to 345 bar.

4.6 NO 15/9 -F-11B IOR, Retrieve SAFELOCK plug11 B, Other (30.06.2013 07:15)

START: 30.Jun.2013 07:15 4770 mMD
END : 30.Jun.2013 14:15 4770 mMD

OBJECTIVE
- Retrieve SAFE LOCK Plug (VMB)

SUMMARY

RIH with Safe Lock plug RT on 5 1/2" DP from surface to 601 m. commenced circulation with 300 liter/min, 3 bar. Latched onto plug and retrieved the plug according ARCHER procedure. Pulled the plug out of hole. Picked up bowl protector RT and retrieved Bowl protector.

4.7 NO 15/9 -F-11B IOR11 B, Run completion, Completion (30.06.2013 14:15)

START: 30.Jun.2013 14:15 4770 mMD
END : 04.Jul.2013 20:00 4770 mMD

OBJECTIVE
- Run in holeRunning the 7” completion string including Mule shoe, Production packer, Down hole gauge and Downhole safety valve.
- Spacestring an dperform space out tubing to tope of liner PBR.

SUMMARY

Rigged up the 7" handling and make up equipment. Completion running started by running the Aassembly#1 (Self Aligning Mule shoe and perforated stem) , Assembly #2 with pre-installed EVO plug and auto fill valve, a 7" tubing joint, Assembly #3 with BOT Granite packer, 2 ea. plain 7" tubing joints. Rigged up the spooling unit for DHPG. Made up the Assembly #4 with DHPG. Connected the Cable to the gauge and tested the connection to 7000 psi. Continue to run with 7" tubing down to 2589m with 6,4 average running speed. Picked up and made up baker DHSV. Installed and terminated DHSV control line. Control line tested to 520 bar. Maintained DHSV open while continue run in hole, at 20,55 meter below rotary observed string taking 15 Ton weight, not able to enter DHSV centralizer into 10 3/4" casing in multi-bowl. Pulled out to inspect DHSV, no mark was on the DHSV. Observed rig not proper alignet above well, skidded rig 15 cm forward. Attempted to run through multi-bowl, still observing string taking weight. Pulled out to inspect DHSV assembly for contact marks, no marks observed. Ran back in with DHSV assembly, sat down 10 - 15 Ton weight on 10 3/4" casing stump, passed through. RIH with 7" tubing to 3066 m with average running speed 5 joints/hr. Made up the space out string and perform space out run. Pulled out and layed out the space out string.

4.8 NO 15/9 -F-11B IOR11 B, Install tubing hanger, Tubing Hanger (04.07.2013 20:00)

START: 04.Jul.2013 20:00 4770 mMD
END : 06.Jul.2013 12:30 4770 mMD

OBJECTIVE
- Make up and land tubing hanger and THRT.
- Terminate lines and install communication collar and hanger running tool.
- Land and leak test the tubing hanger seal void
- Drop EVO plug ball and circulate down with brine until seated
- Set the production packer and pressure test the completion to 345 bar.
- Inflow test the DHSV
- Retrieve EVO plug with tractor. Evaluate if tractor is needed to retrieve plug. (Approx. 62 deg deviation)
- Pressure up annulus to leak test the production packer from above to 345 bar. Inflow test A-annulus inner Valve.
- Release and pull out with THRT

SUMMARY

Picked up and made up tubing hanger to the completion string. Fed the control lines through hanger and terminated. Leak tested all penetrations / fittings to 35 / 450 bar. Installed communication collar according to GE procedure. Made up landing string, ran in hole and landed the Tubing hanger. Pressure tested liner hanger seal to 35 / 345 bar. Rigged up swedge and cement hose. Dropped the ball for activation of EVO plug. Pumped down the ball with 3500 lpm. Pressure increased to 70 bar when the ball landed. Pressure increased to 345 bar and held for 30 min to fully set the packer. Tubing and annulus tested to 345 bar successfully. Rigged up wireline and retrieved the EVO plug. Pressure tested packer from below to 345 bar with 1,03 sg brine. Tubing hanger released and retrieved. BPV set on 3 1/2" DP. Nippled down BOP and retrieved surface BOP riser.

4.9 NO 15/9 -F-11B IOR11 B, Install X-mas treeTree (06.07.2013 12:30)

START: 06.Jul.2013 12:30 4770 mMD
END : 09.Jul.2013 00:45 4770 mMD

OBJECTIVE
- Run in hole. Installation of spacer joint and install BPV plug In Tubing Hanger profile
- N/D BOP/Diverter and Surface Riser
- Install Spacer spool c/w NT-2 connector
- Install and connect the X-mas tree and test of X-mas tree to 345 bar.

SUMMARY

Dart installed in BPV and Terminated control lines on wellhead. Run down with spacer spool and landed on multi-bowl. Made up spacer spool and tested according to GE procedure. X-mas tree lifted and orientation verified. X-mas tree made up to the spacer spool. All valves tested to 345 bar. Well handed over to production for installation of instruments and flow-line.

4.10 NO 15/9 -F-11B IOR11 B, Perforation (09.07.2013 00:45)

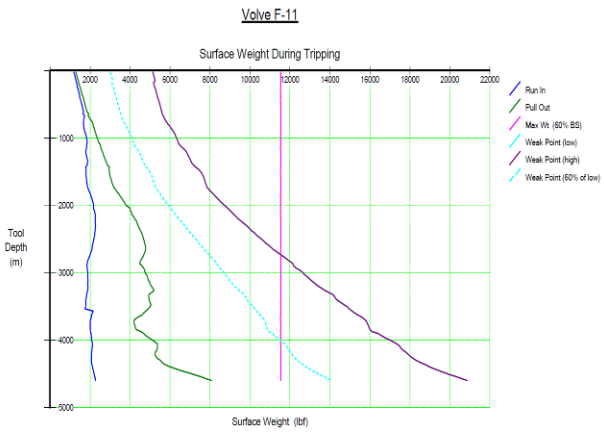
START: 09.Jul.2013 00:45 4770 mMD
END : 21.Jul.2013 23:00 4770 mMD

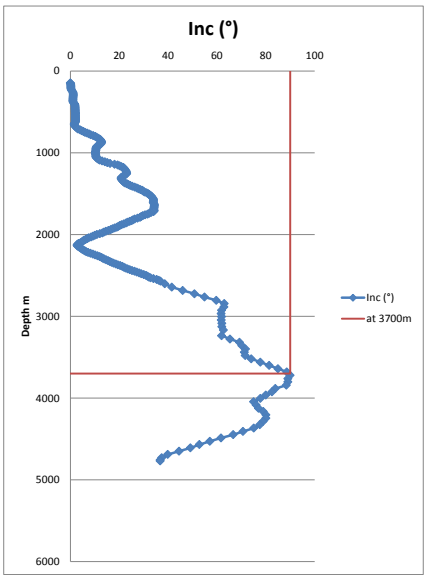
OBJECTIVE
- Rig up WL surface pressure control equipment
- Perforate Well with Wireline/ tractor (8 intervals)
- Rig Down WL- equipment

SUMMARY

Rigged up wire line equipment for perforation operation with PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF guns. Gamma ray tool added to string for the first run to correlate liner. Following perforation runs performed:

Perf#1 (From 4527 to 4539 m MD)
Run no. 1 (Tool length: 29.86m, MAX OD:4.5 in, Tool Weight:1157.65 kg)
(Failed / Tractor problem)
The simulation by Aker well service shows that a 2 sonde tractor (with 1000 lbf power) should be able to cope with the resistance forces; particularly at the depth 3700 m which has the highest angle (90 deg). But BHA was not able to pass the mentioned depth. String was retrieved to surface. No damage or particular scratch was observed on the string.





Run no.2 (Tool length: 29.86m, MAX OD: 4.5 in, Tool Weight: 1157.65 kg)
(Successful / cable snapped at surface)
Back up tool re-run. Same problem was observed at 3700m although after several attempts string passed the problem point. The interval #1 perforated successfully. It was concluded; the actual well friction/resistance is higher than predictions so a new set of tractors with 3 sondes and 1500 lbf power was ordered.
Although the gun was fired successfully, when the BHA reached surface and entered to the lubricator, cable was snapped off. It was parted from the weak point on the cable head. Fortunately tool was caught by tool catcher in

lubricator so nothing was lost in the well nor anything damaged. At the time which the cable was parted the Aker acquisition unit was shut down by accident so no record for tension or over pull was available.

In addition to the probable overpull; the explosion shock could have some contribution to the incident so it was decided to cut the last 20 m of cable and rebuilt cable head after each perforation. Also it should be mentioned that the cable was produced in 2002 and no job/tension history was available.

Perf #2 (From 4488 To 4500 m MD)
(Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
(Successful)
Perf #3 (From 4355 To 4367 m MD)
Run no. 1 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
(Failed / communication was lost)
BHA was run in hole down to 4420 m but communication was lost. Tool was retrieved and laid out for inspection. A short circuit was found in the cable head.

Run no. 2 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
(Successful)

Perf #4 (From 4304 To 4316 m MD)
Run no. 1 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Failed / communication was lost
Run no. 2 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Failed / Cable OD was bigger than Flow tubes
Run no. 3 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Failed / APM or Cable head failed at 3332
First attempt for Perf#4 was failed due to losing communication at 4470m. BHA was pulled out of hole and the cable reel replaced by the new ordered back up (2010 made). After replacement it was found that the new cable strands were too brittle for making cable head. It was probably due to H2S environment in previous jobs. 800 m of cable was slipped and cut to get the acceptable mechanical properties in the QC test.
BHA re-run in hole but due to wrong ID of flow tubes, tool could not pass the 3670m depth. During slip and cut the maximum cable diameter in the active section had increased to 0.423 in but flow tubes with 0.423 in ID were kept on top of lubricator. Tool had to be retrieved and tubes were changed to 0.427in ID.
Third attempt also failed due to communication lost at 3332m. Short circuit was found in cable head. All cable head blocks were replaced by new ordered backups.

Run no. 4 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Successful

Perf# 5 (From 4286 To 4295 m MD)
Run no. 1 (Tool length: 27.01m, MAX OD: 4.5in, Tool Weight: 974.65 kg)
Successful

Perf# 6 (From 4268 To 4280 m MD)
Run no. 1 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Successful

Perf# 7 (From 4063 To 4075 m MD)
Run no. 1 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Successful

Perf# 8 (From 4031 To 4043 m MD)

Run no. 1 (Tool length: 30.85m, MAX OD:4.5 in, Tool Weight: 1184.65 kg)
Successful

Last wire line string pulled out of hole and wire line equipment was rigged down.

4.11 15/9 F-11B IOR, Handover (22.07.2013 14:00) (Offline activity)

START: 22.Jul.2013 14:00 4770 mMD
END : 23.Jul.2013 20:00 4770 mMD

OBJECTIVE
- Test the DHSV and X-mass tree valves prior handing over the well to the production.

SUMMARY

Filled Xmas tree with MEG using MEG injection pump. Pressure tested tree cap to 30/200 bar for 5/10min. Pressure tested Swab valve to 30/200 bar for 5/10 min. Bullheaded 3 m3 MEG into tubing using MEG injection pump, followed by 6,6 m3 packer fluid using cement pump and 1,2 m3 MEG with MEG injection pump. Closed DHSV and inflow tested same for 30 min. Inflow tested the HMV, MMV for 10 m in. Installed cap on KWV and pressure tested KWV cap and KWV to 35/200 bar for 5/10min.
Well F-11B handed over to production 23.07.2013 at 20:00 hrs.

5 Formation evaluation

5.1 Coring summary

N/A

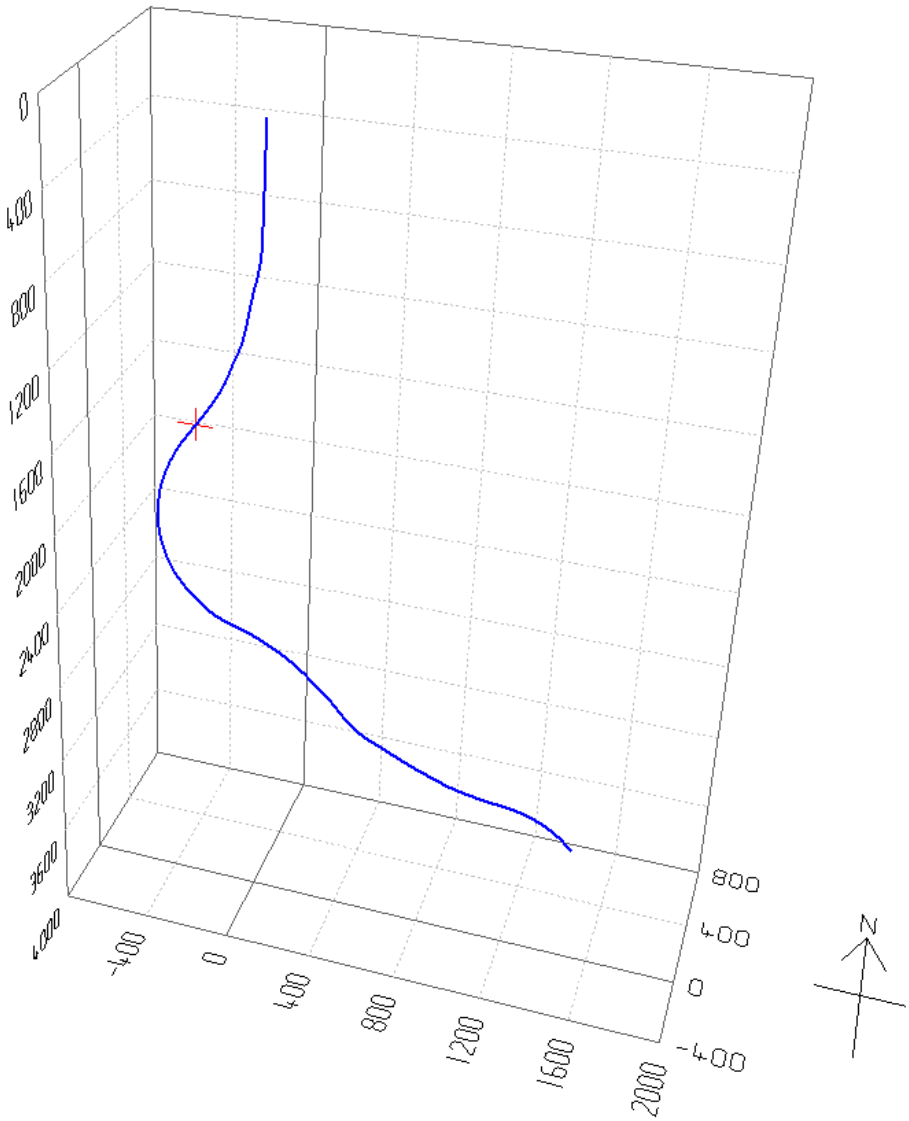
5.2 Pressure points

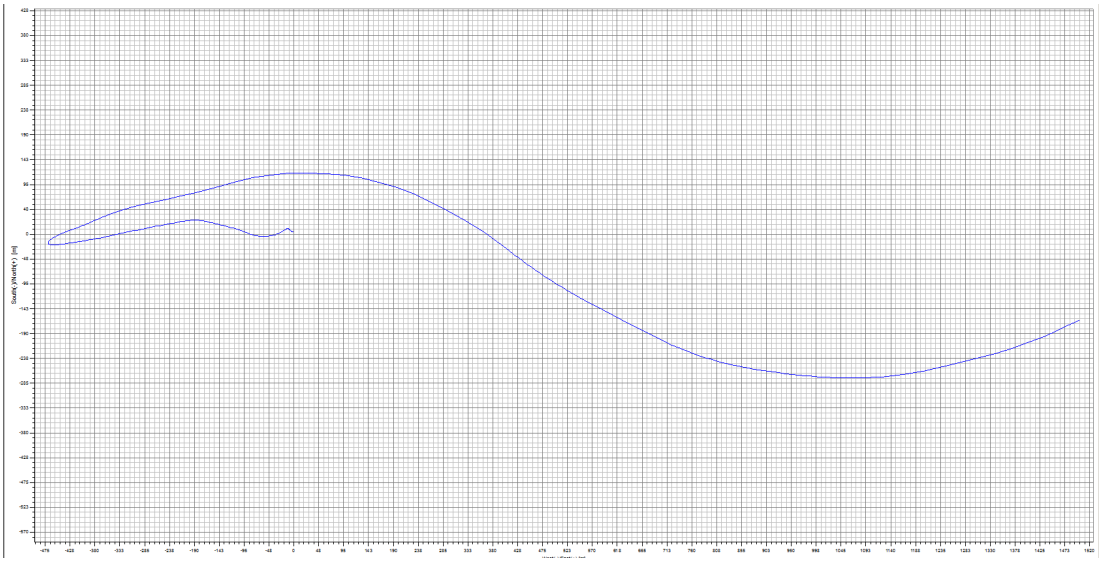
No.	Depth m MD RKB	Depth m TVD RKB	Initial hydrostatic pressure bar	Formation pressure bar	FP gradient g/cc RKB	Final hydrostatic pressure bar	Formation name	Remarks
1	4490,1	3328,5	466,42	367,51	1,13	467,67	Hugin	17 mD/cP; good test
2	4468,3	3307,1	463,97	364,55	1,12	465,14	Hugin	17 mD/cP; good test
3	4451,4	3290,6	461,64	362,78	1,12	462,56	Hugin	124 mD/cP; good test
4	4417,0	3257,6	457,03	361,09	1,13	458,00	Hugin	7 mD/cP; good test
5	4408,9	3249,8	455,61	356,75	1,12	456,52	Hugin	7 mD/cP; good test
6	4395,1	3235,5	453,24	356,67	1,12	454,06	Hugin	22 mD/cP; good test

5.3 Geological formation tops

Formation	Actual m MD RKB	Actual m TVD RKB	Prognosed m MD RKB	Prognosed m TVD RKB	Deviation m TVD
Utsira Fm Top	885	882	884	882	0
Utsira Fm Base	1071	1064	1072	1065	0
Skade Fm Top	1240	1224	1220	1205	19
Skade Fm Base	1330	1308	1338	1310	-2
Grid Fm Top	2221	2119	2180	2078	41
Grid Fm Base	2283	2180	2247	2145	35
Balder Fm Top	2373	2267	2358	2252	15
Sele Fm Top	2435	2324	2420	2309	15
Lista Fm Top	2511	2391	2493	2374	17
Ty Fm Top	2624	2482	2597	2462	21
Ekofisk Fm Top	2787	2590	2789	2590	0
Tor Fm Top	2822	2608	2817	2605	3
Hod Fm Top	3090	2733	3080	2728	5
Blodoeks Fm Top	3219	2793			
Hidra Fm Top	3268	2815	3264	2810	5
Rødby Fm Top	3299	2828	3274	2815	13
Åsgard Fm Top			3291	2821	
Draupne Fm Top (BCU)	3352	2847	3355	2846	0
Heather Fm Top	3352	2847	3368	2851	-5
Heather 'SST' Top			3448	2876	
Hugin Fm Top	3467	2884	3495	2891	-7
Sleipner Fm Top	4732	3227	Not expected -		
TD	4770	3258	4865	3199	

6 Appendix 1: Directional data





Norway

SLEIPNER
Volve F
F-11

15/9-F-11 B

Design: 15/9-F-11 B

Standard Survey Report

09 september, 2013

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Project SLEIPNER, Norway

Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site Volve F, 15/9

Site Position:		Northing:	6 478 563,53 m	Latitude:	58° 26' 29,807 N
From:	Map	Easting:	435 050,02 m	Longitude:	1° 53' 14,929 E
Position Uncertainty:	0,00 m	Slot Radius:	13,200 in	Grid Convergence:	-0,95 °

Well F-11

Well Position	+N/-S	4,65 m	Northing:	6 478 568,18 m	Latitude:	58° 26' 29,957 N
	+E/-W	-0,93 m	Easting:	435 049,09 m	Longitude:	1° 53' 14,867 E
Position Uncertainty		0,00 m	Wellhead Depth:	91,00 m	Water Depth:	91,00 m

Wellbore 15/9-F-11 B

Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2012	27.09.2012	-1,33	71,62	50 495

Design 15/9-F-11 B

Audit Notes:

Version:	1.0	Phase:	ACTUAL	Tie On Depth:	2 570,00

Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145,90	4,65	-0,93	96,40

Survey Program Date 01.07.2013

From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
150,00	2 570,00	F-11 T2 SDI Keeper Gyro (15/9-F-11 T2)	Keeper, cont	Standard Wireline Keeper, Scientific Drilling, (ve
2 603,30	3 167,70	12 1/4"_BH_MWD (15/9-F-11 B)	Magn, IFR, mag-corr, dual in	Magnetic Tools (MWD, EMS)
3 237,80	4 757,10	8 1/2"_BH_MWD_Multistation (15/9-F-11 B	Magnetic, std, non-mag	Magnetic Tools (MWD, EMS)
4 770,00	4 770,00	Projection (15/9-F-11 B)	Blind Drilling	No survey recorded - blind drilling

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
150,00	0,09	227,76	150,00	4,65	-0,93	0,00	0,000	0,000	0,000
160,00	0,20	225,57	160,00	4,63	-0,95	-0,02	0,330	0,330	-6,570
170,00	0,19	227,59	170,00	4,61	-0,97	-0,04	0,036	-0,030	6,060
180,00	0,23	231,32	180,00	4,58	-1,00	-0,07	0,127	0,120	11,190
190,00	0,23	233,02	190,00	4,56	-1,03	-0,09	0,020	0,000	5,100
200,00	0,22	235,12	200,00	4,54	-1,06	-0,12	0,039	-0,030	6,300
210,00	0,24	234,55	210,00	4,51	-1,10	-0,15	0,060	0,060	-1,710
220,00	0,34	239,76	220,00	4,49	-1,14	-0,19	0,310	0,300	15,630

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
230,00	0,42	244,41	230,00	4,46	-1,20	-0,25	0,257	0,240	13,950
240,00	0,57	252,51	240,00	4,42	-1,28	-0,32	0,495	0,450	24,300
250,00	0,92	274,57	250,00	4,42	-1,41	-0,45	1,339	1,050	66,180
260,00	1,13	286,19	260,00	4,45	-1,58	-0,63	0,883	0,630	34,860
270,00	1,16	287,83	269,99	4,51	-1,77	-0,82	0,133	0,090	4,920
280,00	1,20	290,33	279,99	4,58	-1,97	-1,02	0,196	0,120	7,500
290,00	1,24	293,03	289,99	4,65	-2,16	-1,23	0,210	0,120	8,100
300,00	1,20	293,20	299,99	4,74	-2,36	-1,43	0,120	-0,120	0,510
310,00	1,19	294,42	309,99	4,82	-2,55	-1,63	0,082	-0,030	3,660
320,00	1,15	292,81	319,98	4,90	-2,74	-1,83	0,155	-0,120	-4,830
330,00	1,12	290,98	329,98	4,98	-2,92	-2,02	0,141	-0,090	-5,490
340,00	1,10	290,84	339,98	5,05	-3,10	-2,21	0,061	-0,060	-0,420
350,00	1,08	288,68	349,98	5,11	-3,28	-2,39	0,137	-0,060	-6,480
360,00	1,07	287,43	359,98	5,17	-3,46	-2,57	0,076	-0,030	-3,750
370,00	1,10	286,47	369,97	5,22	-3,64	-2,76	0,105	0,090	-2,880
380,00	1,19	289,32	379,97	5,29	-3,83	-2,96	0,319	0,270	8,550
390,00	1,33	294,68	389,97	5,37	-4,03	-3,17	0,549	0,420	16,080
400,00	1,73	303,22	399,97	5,50	-4,27	-3,41	1,378	1,200	25,620
410,00	1,86	305,68	409,96	5,68	-4,52	-3,69	0,453	0,390	7,380
420,00	1,89	306,23	419,96	5,87	-4,79	-3,97	0,105	0,090	1,650
430,00	1,96	306,14	429,95	6,07	-5,06	-4,27	0,210	0,210	-0,270
440,00	1,99	306,93	439,94	6,27	-5,34	-4,56	0,122	0,090	2,370
450,00	2,01	307,01	449,94	6,48	-5,62	-4,86	0,061	0,060	0,240
460,00	1,99	307,09	459,93	6,69	-5,89	-5,16	0,061	-0,060	0,240
470,00	2,03	307,66	469,93	6,91	-6,17	-5,46	0,134	0,120	1,710
480,00	2,06	308,33	479,92	7,13	-6,45	-5,77	0,115	0,090	2,010
490,00	2,05	307,74	489,91	7,35	-6,74	-6,07	0,070	-0,030	-1,770
500,00	2,10	310,18	499,91	7,57	-7,02	-6,38	0,305	0,150	7,320
510,00	2,11	309,44	509,90	7,81	-7,30	-6,69	0,087	0,030	-2,220
520,00	2,12	308,88	519,89	8,04	-7,59	-7,00	0,069	0,030	-1,680
530,00	2,09	308,61	529,89	8,27	-7,87	-7,31	0,095	-0,090	-0,810
540,00	2,13	310,13	539,88	8,51	-8,16	-7,62	0,206	0,120	4,560
550,00	2,13	309,28	549,87	8,74	-8,44	-7,93	0,095	0,000	-2,550
560,00	2,12	309,17	559,87	8,98	-8,73	-8,24	0,032	-0,030	-0,330
570,00	2,13	309,03	569,86	9,21	-9,02	-8,55	0,034	0,030	-0,420
580,00	2,16	309,02	579,85	9,45	-9,31	-8,86	0,090	0,090	-0,030
590,00	2,14	309,85	589,85	9,69	-9,60	-9,18	0,111	-0,060	2,490
600,00	2,18	309,02	599,84	9,93	-9,89	-9,50	0,152	0,120	-2,490
610,00	2,15	308,45	609,83	10,16	-10,18	-9,81	0,111	-0,090	-1,710
620,00	2,13	307,49	619,82	10,39	-10,48	-10,13	0,123	-0,060	-2,880
630,00	2,01	302,74	629,82	10,60	-10,77	-10,45	0,628	-0,360	-14,250
640,00	1,68	284,45	639,81	10,73	-11,06	-10,75	2,012	-0,990	-54,870
650,00	1,63	275,56	649,81	10,78	-11,35	-11,04	0,784	-0,150	-26,670

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
660,00	1,66	272,60	659,80	10,80	-11,63	-11,33	0,270	0,090	-8,880
670,00	1,79	276,80	669,80	10,83	-11,93	-11,63	0,544	0,390	12,600
680,00	2,11	265,05	679,79	10,83	-12,27	-11,96	1,532	0,960	-35,250
690,00	2,45	259,38	689,79	10,77	-12,66	-12,35	1,223	1,020	-17,010
700,00	2,77	256,59	699,78	10,68	-13,11	-12,78	1,033	0,960	-8,370
710,00	3,19	251,91	709,76	10,54	-13,61	-13,26	1,455	1,260	-14,040
720,00	4,01	244,94	719,74	10,30	-14,19	-13,81	2,784	2,460	-20,910
730,00	4,67	241,67	729,71	9,96	-14,87	-14,44	2,114	1,980	-9,810
740,00	5,38	239,18	739,67	9,53	-15,63	-15,15	2,228	2,130	-7,470
750,00	6,03	237,66	749,63	9,01	-16,47	-15,93	2,002	1,950	-4,560
760,00	6,89	236,29	759,56	8,39	-17,42	-16,80	2,621	2,580	-4,110
770,00	7,68	235,54	769,48	7,68	-18,47	-17,77	2,387	2,370	-2,250
780,00	8,37	235,75	779,38	6,89	-19,62	-18,83	2,072	2,070	0,630
790,00	9,33	235,86	789,26	6,03	-20,89	-19,99	2,880	2,880	0,330
800,00	10,21	236,82	799,12	5,09	-22,30	-21,29	2,685	2,640	2,880
810,00	10,79	237,99	808,95	4,11	-23,84	-22,71	1,854	1,740	3,510
820,00	11,33	239,39	818,77	3,11	-25,48	-24,23	1,809	1,620	4,200
830,00	11,93	242,49	828,56	2,13	-27,24	-25,87	2,598	1,800	9,300
840,00	12,31	246,06	838,34	1,22	-29,13	-27,65	2,521	1,140	10,710
850,00	12,52	249,24	848,10	0,41	-31,12	-29,53	2,145	0,630	9,540
860,00	12,79	252,71	857,86	-0,31	-33,19	-31,51	2,420	0,810	10,410
870,00	12,84	255,52	867,61	-0,91	-35,32	-33,56	1,876	0,150	8,430
880,00	12,75	256,87	877,36	-1,44	-37,47	-35,64	0,937	-0,270	4,050
890,00	12,55	257,29	887,12	-1,93	-39,61	-37,71	0,660	-0,600	1,260
900,00	12,18	258,31	896,89	-2,38	-41,70	-39,74	1,289	-1,110	3,060
910,00	11,40	260,19	906,68	-2,77	-43,71	-41,69	2,608	-2,340	5,640
920,00	10,91	261,87	916,49	-3,07	-45,62	-43,55	1,764	-1,470	5,040
930,00	10,76	262,94	926,31	-3,32	-47,48	-45,38	0,753	-0,450	3,210
940,00	10,58	264,05	936,14	-3,53	-49,32	-47,18	0,820	-0,540	3,330
950,00	10,53	264,50	945,97	-3,71	-51,14	-48,97	0,289	-0,150	1,350
960,00	10,52	265,36	955,80	-3,87	-52,96	-50,76	0,472	-0,030	2,580
970,00	10,48	266,01	965,63	-4,01	-54,78	-52,55	0,375	-0,120	1,950
980,00	10,36	267,69	975,47	-4,11	-56,59	-54,34	0,980	-0,360	5,040
990,00	10,31	270,33	985,31	-4,14	-58,38	-56,12	1,429	-0,150	7,920
1 000,00	10,29	272,52	995,14	-4,09	-60,17	-57,90	1,176	-0,060	6,570
1 010,00	10,34	273,32	1 004,98	-4,00	-61,95	-59,68	0,455	0,150	2,400
1 020,00	10,39	274,63	1 014,82	-3,88	-63,75	-61,48	0,723	0,150	3,930
1 030,00	10,48	275,34	1 024,65	-3,72	-65,55	-63,29	0,471	0,270	2,130
1 040,00	10,47	275,48	1 034,49	-3,55	-67,36	-65,11	0,082	-0,030	0,420
1 050,00	10,49	275,78	1 044,32	-3,37	-69,17	-66,93	0,174	0,060	0,900
1 060,00	10,66	277,07	1 054,15	-3,17	-71,00	-68,76	0,874	0,510	3,870
1 070,00	11,01	278,92	1 063,97	-2,90	-72,86	-70,64	1,480	1,050	5,550
1 080,00	11,46	281,15	1 073,78	-2,56	-74,78	-72,59	1,876	1,350	6,690
1 090,00	12,02	283,68	1 083,57	-2,13	-76,76	-74,61	2,282	1,680	7,590

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1 100,00	13,04	287,32	1 093,33	-1,54	-78,85	-76,75	3,869	3,060	10,920
1 110,00	14,19	289,87	1 103,05	-0,79	-81,08	-79,05	3,891	3,450	7,650
1 120,00	15,14	291,09	1 112,73	0,10	-83,45	-81,50	2,997	2,850	3,660
1 130,00	16,29	291,76	1 122,35	1,09	-85,97	-84,12	3,493	3,450	2,010
1 140,00	18,12	291,94	1 131,91	2,19	-88,72	-86,97	5,492	5,490	0,540
1 150,00	19,37	291,62	1 141,37	3,38	-91,70	-90,07	3,763	3,750	-0,960
1 160,00	20,20	291,03	1 150,78	4,61	-94,86	-93,34	2,561	2,490	-1,770
1 170,00	20,94	290,28	1 160,15	5,85	-98,14	-96,75	2,357	2,220	-2,250
1 180,00	21,74	288,95	1 169,46	7,07	-101,57	-100,29	2,805	2,400	-3,990
1 190,00	22,06	288,24	1 178,74	8,26	-105,11	-103,93	1,246	0,960	-2,130
1 200,00	22,30	287,56	1 188,00	9,42	-108,70	-107,63	1,054	0,720	-2,040
1 210,00	22,53	286,61	1 197,24	10,54	-112,34	-111,38	1,287	0,690	-2,850
1 220,00	22,78	285,44	1 206,47	11,60	-116,05	-115,18	1,546	0,750	-3,510
1 230,00	22,91	284,77	1 215,69	12,61	-119,79	-119,01	0,872	0,390	-2,010
1 240,00	22,99	284,46	1 224,90	13,60	-123,57	-122,87	0,435	0,240	-0,930
1 250,00	23,10	284,53	1 234,10	14,58	-127,36	-126,75	0,340	0,330	0,210
1 260,00	23,02	284,46	1 243,30	15,56	-131,15	-130,63	0,254	-0,240	-0,210
1 270,00	22,40	284,23	1 252,52	16,51	-134,89	-134,45	1,879	-1,860	-0,690
1 280,00	22,06	284,22	1 261,78	17,44	-138,56	-138,20	1,020	-1,020	-0,030
1 290,00	21,49	284,19	1 271,07	18,35	-142,15	-141,87	1,710	-1,710	-0,090
1 300,00	20,95	283,85	1 280,39	19,23	-145,66	-145,46	1,662	-1,620	-1,020
1 310,00	20,83	283,60	1 289,73	20,08	-149,13	-149,00	0,448	-0,360	-0,750
1 320,00	20,85	283,68	1 299,08	20,92	-152,59	-152,53	0,104	0,060	0,240
1 330,00	21,02	283,73	1 308,42	21,76	-156,06	-156,07	0,513	0,510	0,150
1 340,00	21,31	283,42	1 317,74	22,61	-159,57	-159,65	0,933	0,870	-0,930
1 350,00	21,61	283,17	1 327,05	23,45	-163,13	-163,29	0,941	0,900	-0,750
1 360,00	21,97	282,58	1 336,33	24,28	-166,75	-166,97	1,264	1,080	-1,770
1 370,00	22,45	281,18	1 345,59	25,06	-170,44	-170,74	2,143	1,440	-4,200
1 380,00	23,22	278,29	1 354,81	25,71	-174,27	-174,61	4,081	2,310	-8,670
1 390,00	23,84	276,25	1 363,98	26,21	-178,23	-178,60	3,070	1,860	-6,120
1 400,00	24,53	274,57	1 373,10	26,60	-182,31	-182,70	2,924	2,070	-5,040
1 410,00	25,42	271,77	1 382,17	26,83	-186,52	-186,91	4,439	2,670	-8,400
1 420,00	26,38	269,11	1 391,16	26,86	-190,89	-191,25	4,521	2,880	-7,980
1 430,00	27,21	266,65	1 400,09	26,70	-195,39	-195,71	4,155	2,490	-7,380
1 440,00	27,94	264,45	1 408,95	26,34	-200,01	-200,26	3,759	2,190	-6,600
1 450,00	28,74	261,88	1 417,75	25,77	-204,72	-204,88	4,376	2,400	-7,710
1 460,00	29,48	260,21	1 426,49	25,01	-209,52	-209,57	3,297	2,220	-5,010
1 470,00	29,92	259,85	1 435,18	24,15	-214,40	-214,32	1,424	1,320	-1,080
1 480,00	30,40	259,60	1 443,82	23,26	-219,35	-219,13	1,488	1,440	-0,750
1 490,00	31,22	259,49	1 452,41	22,33	-224,38	-224,04	2,466	2,460	-0,330
1 500,00	31,87	259,45	1 460,93	21,37	-229,53	-229,04	1,951	1,950	-0,120
1 510,00	32,35	259,26	1 469,40	20,39	-234,75	-234,12	1,472	1,440	-0,570
1 520,00	32,72	259,31	1 477,84	19,39	-240,04	-239,26	1,113	1,110	0,150

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1 530,00	33,21	259,32	1 486,23	18,38	-245,38	-244,46	1,470	1,470	0,030
1 540,00	33,41	259,40	1 494,58	17,37	-250,78	-249,71	0,614	0,600	0,240
1 550,00	33,61	259,49	1 502,92	16,36	-256,21	-255,00	0,618	0,600	0,270
1 560,00	33,95	259,53	1 511,23	15,34	-261,67	-260,32	1,022	1,020	0,120
1 570,00	34,05	259,37	1 519,52	14,32	-267,17	-265,66	0,403	0,300	-0,480
1 580,00	34,02	258,98	1 527,81	13,27	-272,67	-271,01	0,661	-0,090	-1,170
1 590,00	34,03	258,75	1 536,10	12,19	-278,16	-276,35	0,387	0,030	-0,690
1 600,00	33,97	258,48	1 544,39	11,08	-283,64	-281,67	0,487	-0,180	-0,810
1 610,00	34,00	258,45	1 552,68	9,97	-289,12	-286,99	0,103	0,090	-0,090
1 620,00	34,20	258,47	1 560,96	8,84	-294,61	-292,32	0,601	0,600	0,060
1 630,00	34,38	258,36	1 569,22	7,71	-300,13	-297,68	0,571	0,540	-0,330
1 640,00	34,44	258,17	1 577,47	6,56	-305,66	-303,05	0,369	0,180	-0,570
1 650,00	34,42	257,88	1 585,72	5,39	-311,19	-308,42	0,496	-0,060	-0,870
1 660,00	34,35	257,77	1 593,97	4,20	-316,71	-313,77	0,281	-0,210	-0,330
1 670,00	34,31	257,67	1 602,23	3,00	-322,23	-319,11	0,207	-0,120	-0,300
1 680,00	34,23	257,70	1 610,49	1,80	-327,73	-324,45	0,245	-0,240	0,090
1 690,00	34,23	257,66	1 618,76	0,60	-333,22	-329,78	0,068	0,000	-0,120
1 700,00	34,30	257,68	1 627,03	-0,60	-338,72	-335,11	0,213	0,210	0,060
1 710,00	34,41	257,59	1 635,28	-1,81	-344,23	-340,45	0,363	0,330	-0,270
1 720,00	34,04	257,50	1 643,55	-3,03	-349,73	-345,77	1,120	-1,110	-0,270
1 730,00	33,09	257,75	1 651,88	-4,21	-355,13	-351,01	2,880	-2,850	0,750
1 740,00	32,35	258,27	1 660,30	-5,33	-360,41	-356,14	2,375	-2,220	1,560
1 750,00	31,64	258,71	1 668,78	-6,39	-365,61	-361,18	2,242	-2,130	1,320
1 760,00	30,59	259,30	1 677,34	-7,38	-370,68	-366,11	3,280	-3,150	1,770
1 770,00	29,68	259,79	1 685,99	-8,29	-375,62	-370,91	2,828	-2,730	1,470
1 780,00	29,07	260,04	1 694,70	-9,15	-380,44	-375,62	1,867	-1,830	0,750
1 790,00	28,52	260,12	1 703,46	-9,98	-385,19	-380,24	1,654	-1,650	0,240
1 800,00	27,54	260,26	1 712,29	-10,78	-389,82	-384,75	2,947	-2,940	0,420
1 810,00	26,37	260,65	1 721,21	-11,53	-394,29	-389,11	3,550	-3,510	1,170
1 820,00	25,69	260,79	1 730,19	-12,24	-398,62	-393,34	2,048	-2,040	0,420
1 830,00	25,10	261,16	1 739,22	-12,91	-402,86	-397,47	1,833	-1,770	1,110
1 840,00	24,31	261,30	1 748,31	-13,55	-406,99	-401,50	2,376	-2,370	0,420
1 850,00	23,40	261,24	1 757,45	-14,16	-410,98	-405,41	2,731	-2,730	-0,180
1 860,00	22,79	260,74	1 766,65	-14,78	-414,86	-409,19	1,922	-1,830	-1,500
1 870,00	21,92	260,49	1 775,90	-15,40	-418,61	-412,85	2,626	-2,610	-0,750
1 880,00	20,98	260,58	1 785,21	-16,00	-422,22	-416,37	2,822	-2,820	0,270
1 890,00	20,23	261,02	1 794,57	-16,56	-425,69	-419,76	2,297	-2,250	1,320
1 900,00	19,65	261,39	1 803,97	-17,08	-429,06	-423,05	1,781	-1,740	1,110
1 910,00	19,02	261,79	1 813,41	-17,57	-432,34	-426,25	1,931	-1,890	1,200
1 920,00	18,16	262,11	1 822,88	-18,01	-435,49	-429,34	2,598	-2,580	0,960
1 930,00	17,32	262,45	1 832,41	-18,42	-438,51	-432,29	2,539	-2,520	1,020
1 940,00	16,56	262,81	1 841,97	-18,80	-441,40	-435,12	2,302	-2,280	1,080
1 950,00	15,62	263,53	1 851,58	-19,13	-444,15	-437,82	2,883	-2,820	2,160

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
1 960,00	14,58	264,78	1 861,24	-19,39	-446,75	-440,36	3,269	-3,120	3,750
1 970,00	13,75	265,49	1 870,93	-19,60	-449,18	-442,76	2,544	-2,490	2,130
1 980,00	13,12	265,75	1 880,66	-19,78	-451,50	-445,05	1,899	-1,890	0,780
1 990,00	12,11	266,00	1 890,42	-19,94	-453,68	-447,19	3,034	-3,030	0,750
2 000,00	10,90	266,44	1 900,22	-20,07	-455,67	-449,16	3,640	-3,630	1,320
2 010,00	10,04	267,49	1 910,05	-20,16	-457,48	-450,95	2,643	-2,580	3,150
2 020,00	9,28	269,00	1 919,91	-20,22	-459,16	-452,61	2,403	-2,280	4,530
2 030,00	8,48	271,02	1 929,79	-20,22	-460,70	-454,14	2,576	-2,400	6,060
2 040,00	7,52	273,53	1 939,69	-20,16	-462,09	-455,53	3,064	-2,880	7,530
2 050,00	6,87	275,05	1 949,61	-20,07	-463,34	-456,78	2,032	-1,950	4,560
2 060,00	6,28	277,53	1 959,55	-19,95	-464,48	-457,93	1,964	-1,770	7,440
2 070,00	5,66	282,49	1 969,49	-19,77	-465,50	-458,96	2,418	-1,860	14,880
2 080,00	5,24	287,86	1 979,45	-19,52	-466,42	-459,90	1,981	-1,260	16,110
2 090,00	4,77	293,02	1 989,41	-19,22	-467,24	-460,75	1,951	-1,410	15,480
2 100,00	4,19	297,86	1 999,38	-18,89	-467,94	-461,49	2,076	-1,740	14,520
2 110,00	3,37	306,50	2 009,36	-18,54	-468,50	-462,08	2,989	-2,460	25,920
2 120,00	2,89	317,71	2 019,34	-18,18	-468,91	-462,52	2,327	-1,440	33,630
2 130,00	2,79	328,96	2 029,33	-17,78	-469,20	-462,86	1,696	-0,300	33,750
2 140,00	2,93	341,18	2 039,32	-17,33	-469,41	-463,12	1,873	0,420	36,660
2 150,00	3,39	355,71	2 049,30	-16,80	-469,52	-463,28	2,760	1,380	43,590
2 160,00	3,94	7,56	2 059,28	-16,16	-469,49	-463,33	2,800	1,650	35,550
2 170,00	4,35	16,27	2 069,26	-15,46	-469,34	-463,26	2,250	1,230	26,130
2 180,00	4,85	25,12	2 079,22	-14,71	-469,06	-463,06	2,600	1,500	26,550
2 190,00	5,41	33,13	2 089,18	-13,93	-468,62	-462,71	2,724	1,680	24,030
2 200,00	6,00	38,53	2 099,13	-13,13	-468,04	-462,22	2,391	1,770	16,200
2 210,00	6,56	42,36	2 109,07	-12,30	-467,33	-461,61	2,097	1,680	11,490
2 220,00	7,30	46,42	2 119,00	-11,44	-466,48	-460,86	2,661	2,220	12,180
2 230,00	8,25	51,54	2 128,91	-10,55	-465,46	-459,95	3,524	2,850	15,360
2 240,00	9,24	55,20	2 138,79	-9,65	-464,24	-458,83	3,406	2,970	10,980
2 250,00	10,14	57,43	2 148,65	-8,72	-462,84	-457,54	2,925	2,700	6,690
2 260,00	11,11	59,85	2 158,48	-7,76	-461,26	-456,09	3,202	2,910	7,260
2 270,00	12,04	61,78	2 168,27	-6,78	-459,51	-454,45	3,022	2,790	5,790
2 280,00	12,77	63,25	2 178,04	-5,79	-457,60	-452,67	2,386	2,190	4,410
2 290,00	13,29	64,43	2 187,78	-4,80	-455,58	-450,77	1,752	1,560	3,540
2 300,00	13,92	65,47	2 197,50	-3,80	-453,45	-448,76	2,027	1,890	3,120
2 310,00	14,69	65,98	2 207,19	-2,79	-451,20	-446,64	2,341	2,310	1,530
2 320,00	15,47	66,14	2 216,85	-1,73	-448,82	-444,39	2,343	2,340	0,480
2 330,00	16,18	66,32	2 226,47	-0,63	-446,32	-442,03	2,135	2,130	0,540
2 340,00	16,93	66,49	2 236,06	0,51	-443,71	-439,57	2,255	2,250	0,510
2 350,00	17,72	66,96	2 245,60	1,69	-440,98	-436,98	2,407	2,370	1,410
2 360,00	18,61	67,36	2 255,10	2,90	-438,10	-434,26	2,696	2,670	1,200
2 370,00	19,52	67,72	2 264,55	4,14	-435,08	-431,40	2,753	2,730	1,080
2 380,00	20,43	68,12	2 273,95	5,43	-431,92	-428,39	2,761	2,730	1,200
2 390,00	21,25	68,32	2 283,30	6,75	-428,61	-425,26	2,469	2,460	0,600

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2 400,00	21,92	68,35	2 292,60	8,10	-425,20	-422,01	2,010	2,010	0,090
2 410,00	22,61	68,54	2 301,85	9,50	-421,67	-418,67	2,081	2,070	0,570
2 420,00	23,43	68,67	2 311,05	10,92	-418,03	-415,21	2,465	2,460	0,390
2 430,00	24,36	68,72	2 320,20	12,39	-414,26	-411,62	2,791	2,790	0,150
2 440,00	25,31	68,55	2 329,27	13,92	-410,35	-407,90	2,858	2,850	-0,510
2 450,00	26,29	68,14	2 338,28	15,53	-406,30	-404,06	2,988	2,940	-1,230
2 460,00	27,23	67,77	2 347,20	17,22	-402,13	-400,10	2,864	2,820	-1,110
2 470,00	28,23	67,35	2 356,06	19,00	-397,83	-396,03	3,057	3,000	-1,260
2 480,00	29,08	67,17	2 364,83	20,85	-393,40	-391,84	2,563	2,550	-0,540
2 490,00	30,08	67,26	2 373,53	22,76	-388,85	-387,53	3,003	3,000	0,270
2 500,00	31,02	67,18	2 382,14	24,73	-384,17	-383,09	2,823	2,820	-0,240
2 510,00	31,64	67,35	2 390,68	26,74	-379,37	-378,55	1,879	1,860	0,510
2 520,00	32,15	67,45	2 399,17	28,77	-374,49	-373,93	1,538	1,530	0,300
2 530,00	32,87	67,45	2 407,60	30,83	-369,53	-369,23	2,160	2,160	0,000
2 540,00	34,06	67,96	2 415,95	32,92	-364,43	-364,39	3,668	3,570	1,530
2 550,00	35,36	68,34	2 424,17	35,04	-359,14	-359,37	3,954	3,900	1,140
2 560,00	36,24	68,58	2 432,28	37,19	-353,70	-354,20	2,673	2,640	0,720
2 570,00	36,93	68,78	2 440,31	39,36	-348,15	-348,93	2,101	2,070	0,600
2 580,00	37,45	69,93	2 448,27	41,49	-342,49	-343,54	2,595	1,548	3,445
2 590,00	37,97	71,05	2 456,19	43,53	-336,73	-338,04	2,595	1,581	3,364
2 600,00	38,51	72,14	2 464,04	45,48	-330,85	-332,42	2,595	1,612	3,285
2 610,00	39,16	73,01	2 471,83	47,36	-324,87	-326,69	2,549	1,956	2,604
2 620,00	39,87	73,76	2 479,54	49,18	-318,78	-320,83	2,561	2,128	2,238
2 630,00	40,59	74,48	2 487,18	50,94	-312,56	-314,85	2,561	2,142	2,172
2 640,00	41,30	75,19	2 494,73	52,66	-306,24	-308,76	2,561	2,156	2,109
2 650,00	42,27	75,78	2 502,19	54,32	-299,79	-302,54	3,145	2,908	1,797
2 660,00	43,37	76,32	2 509,53	55,96	-293,19	-296,17	3,473	3,298	1,601
2 670,00	44,48	76,83	2 516,73	57,57	-286,45	-289,64	3,473	3,305	1,537
2 680,00	45,58	77,32	2 523,80	59,15	-279,55	-282,96	3,473	3,312	1,478
2 690,00	46,74	77,61	2 530,73	60,71	-272,51	-276,14	3,526	3,470	0,865
2 700,00	47,93	77,77	2 537,50	62,28	-265,33	-269,18	3,587	3,570	0,482
2 710,00	49,12	77,93	2 544,13	63,86	-258,00	-262,07	3,587	3,570	0,464
2 720,00	50,31	78,08	2 550,59	65,44	-250,54	-254,84	3,587	3,571	0,448
2 730,00	51,40	78,02	2 556,90	67,04	-242,95	-247,47	3,299	3,296	-0,162
2 740,00	52,43	77,82	2 563,07	68,69	-235,26	-240,01	3,119	3,081	-0,618
2 750,00	53,46	77,62	2 569,10	70,39	-227,46	-232,45	3,119	3,082	-0,602
2 760,00	54,49	77,42	2 574,98	72,14	-219,56	-224,79	3,119	3,083	-0,586
2 770,00	55,59	77,12	2 580,71	73,94	-211,57	-217,05	3,402	3,323	-0,892
2 780,00	56,77	76,74	2 586,28	75,82	-203,48	-209,22	3,664	3,537	-1,148
2 790,00	57,95	76,37	2 591,67	77,78	-195,29	-201,30	3,664	3,541	-1,117
2 800,00	59,14	76,01	2 596,89	79,81	-187,00	-193,29	3,664	3,544	-1,089
2 810,00	60,15	75,58	2 601,93	81,93	-178,63	-185,21	3,233	3,043	-1,268
2 820,00	60,98	75,09	2 606,85	84,13	-170,21	-177,08	2,808	2,501	-1,466

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
2 830,00	61,82	74,61	2 611,64	86,43	-161,73	-168,92	2,808	2,507	-1,442
2 840,00	62,66	74,14	2 616,29	88,81	-153,21	-160,71	2,808	2,511	-1,420
2 850,00	63,04	73,92	2 620,84	91,27	-144,65	-152,48	1,283	1,139	-0,664
2 860,00	63,01	73,92	2 625,38	93,74	-136,09	-144,25	0,081	-0,081	-0,007
2 870,00	62,98	73,91	2 629,92	96,21	-127,53	-136,01	0,081	-0,081	-0,007
2 880,00	62,96	73,91	2 634,46	98,68	-118,97	-127,78	0,081	-0,081	-0,007
2 890,00	62,82	74,47	2 639,02	101,13	-110,41	-119,55	1,539	-0,398	1,670
2 900,00	62,56	75,74	2 643,60	103,41	-101,82	-111,27	3,480	-0,776	3,817
2 910,00	62,32	77,02	2 648,23	105,50	-93,21	-102,94	3,480	-0,742	3,835
2 920,00	62,08	78,30	2 652,89	107,39	-84,56	-94,57	3,480	-0,706	3,852
2 930,00	61,92	79,57	2 657,59	109,08	-75,90	-86,14	3,400	-0,469	3,813
2 940,00	61,87	80,82	2 662,30	110,58	-67,21	-77,67	3,307	-0,156	3,745
2 950,00	61,83	82,07	2 667,02	111,89	-58,49	-69,15	3,307	-0,122	3,748
2 960,00	61,80	83,32	2 671,74	113,02	-49,75	-60,59	3,307	-0,088	3,751
2 970,00	61,80	84,34	2 676,47	113,95	-40,98	-51,99	2,689	-0,004	3,051
2 980,00	61,83	84,96	2 681,19	114,77	-32,21	-43,36	1,638	0,099	1,855
2 990,00	61,87	85,58	2 685,91	115,50	-23,42	-34,71	1,638	0,107	1,854
3 000,00	61,91	86,19	2 690,62	116,13	-14,62	-26,03	1,638	0,115	1,853
3 010,00	61,92	86,98	2 695,33	116,67	-5,82	-17,34	2,091	0,022	2,370
3 020,00	61,89	87,97	2 700,04	117,05	3,00	-8,63	2,608	-0,074	2,955
3 030,00	61,87	88,95	2 704,75	117,29	11,81	0,11	2,608	-0,053	2,956
3 040,00	61,86	89,94	2 709,47	117,38	20,63	8,86	2,608	-0,032	2,957
3 050,00	61,88	90,85	2 714,18	117,31	29,45	17,63	2,413	0,048	2,735
3 060,00	61,93	91,66	2 718,89	117,12	38,27	26,42	2,135	0,148	2,415
3 070,00	61,98	92,46	2 723,59	116,80	47,09	35,22	2,135	0,162	2,412
3 080,00	62,04	93,26	2 728,28	116,36	55,91	44,03	2,135	0,176	2,410
3 090,00	62,08	94,07	2 732,97	115,80	64,72	52,86	2,150	0,106	2,431
3 100,00	62,07	94,90	2 737,65	115,11	73,53	61,69	2,182	-0,017	2,469
3 110,00	62,07	95,72	2 742,34	114,29	82,33	70,52	2,182	-0,002	2,469
3 120,00	62,07	96,54	2 747,02	113,35	91,12	79,36	2,182	0,012	2,469
3 130,00	62,12	97,41	2 751,70	112,28	99,89	88,19	2,313	0,151	2,611
3 140,00	62,27	98,39	2 756,37	111,06	108,65	97,03	2,623	0,446	2,922
3 150,00	62,43	99,36	2 761,01	109,69	117,40	105,88	2,623	0,466	2,914
3 160,00	62,59	100,33	2 765,62	108,18	126,14	114,74	2,623	0,486	2,905
3 170,00	62,69	101,32	2 770,21	106,51	134,87	123,60	2,665	0,294	2,982
3 180,00	62,56	102,41	2 774,81	104,68	143,56	132,44	2,928	-0,392	3,268
3 190,00	62,44	103,50	2 779,43	102,70	152,20	141,25	2,928	-0,367	3,275
3 200,00	62,32	104,60	2 784,07	100,55	160,79	150,03	2,928	-0,341	3,283
3 210,00	62,22	105,69	2 788,72	98,23	169,34	158,78	2,928	-0,315	3,289
3 220,00	62,12	106,79	2 793,39	95,76	177,83	167,49	2,928	-0,289	3,295
3 230,00	62,03	107,89	2 798,07	93,13	186,26	176,16	2,928	-0,263	3,301
3 240,00	62,15	109,04	2 802,77	90,33	194,64	184,80	3,056	0,357	3,435
3 250,00	62,98	110,32	2 807,37	87,34	203,00	193,44	4,238	2,497	3,858

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3 260,00	63,83	111,59	2 811,85	84,15	211,35	202,09	4,238	2,532	3,801
3 270,00	64,68	112,84	2 816,19	80,74	219,68	210,76	4,238	2,564	3,746
3 280,00	65,56	113,98	2 820,40	77,13	228,01	219,43	4,087	2,642	3,437
3 290,00	66,53	114,70	2 824,46	73,36	236,33	228,13	3,492	2,890	2,145
3 300,00	67,49	115,40	2 828,37	69,47	244,67	236,85	3,492	2,900	2,114
3 310,00	68,46	116,10	2 832,12	65,44	253,02	245,59	3,492	2,909	2,085
3 320,00	69,30	116,68	2 835,71	61,28	261,38	254,36	3,002	2,519	1,751
3 330,00	69,54	116,80	2 839,23	57,07	269,74	263,14	0,780	0,709	0,345
3 340,00	69,78	116,91	2 842,70	52,84	278,10	271,92	0,780	0,710	0,344
3 350,00	70,01	117,03	2 846,14	48,58	286,47	280,72	0,780	0,710	0,343
3 360,00	70,26	117,18	2 849,54	44,30	294,85	289,51	0,866	0,741	0,477
3 370,00	70,62	117,86	2 852,89	39,94	303,20	298,30	2,201	1,096	2,025
3 380,00	70,99	118,53	2 856,17	35,48	311,53	307,07	2,201	1,103	2,016
3 390,00	71,36	119,20	2 859,40	30,91	319,82	315,82	2,201	1,111	2,007
3 400,00	71,72	119,88	2 862,56	26,23	328,07	324,54	2,212	1,059	2,048
3 410,00	71,60	120,96	2 865,71	21,43	336,26	333,21	3,081	-0,341	3,226
3 420,00	71,49	122,04	2 868,88	16,47	344,34	341,80	3,081	-0,323	3,230
3 430,00	71,39	123,11	2 872,06	11,37	352,33	350,31	3,081	-0,304	3,234
3 440,00	71,30	124,19	2 875,26	6,12	360,22	358,73	3,073	-0,271	3,230
3 450,00	71,38	125,19	2 878,46	0,73	368,01	367,07	2,842	0,230	2,989
3 460,00	71,46	126,18	2 881,64	-4,80	375,71	375,34	2,842	0,246	2,987
3 470,00	71,55	127,18	2 884,82	-10,47	383,31	383,53	2,842	0,261	2,984
3 480,00	71,64	128,17	2 887,97	-16,27	390,82	391,64	2,842	0,277	2,981
3 490,00	72,24	127,77	2 891,07	-22,11	398,32	399,74	2,140	1,804	-1,212
3 500,00	72,84	127,36	2 894,07	-27,93	405,88	407,90	2,140	1,806	-1,204
3 510,00	73,45	126,97	2 896,97	-33,71	413,51	416,13	2,140	1,809	-1,196
3 520,00	74,05	126,57	2 899,77	-39,46	421,20	424,41	2,140	1,811	-1,189
3 530,00	74,96	126,24	2 902,44	-45,18	428,95	432,75	2,893	2,728	-1,000
3 540,00	75,88	125,91	2 904,96	-50,88	436,78	441,16	2,935	2,777	-0,982
3 550,00	76,81	125,58	2 907,32	-56,55	444,66	449,63	2,935	2,778	-0,974
3 560,00	77,74	125,26	2 909,52	-62,21	452,61	458,16	2,935	2,780	-0,967
3 570,00	78,63	124,83	2 911,57	-67,83	460,62	466,75	2,955	2,675	-1,283
3 580,00	79,52	124,40	2 913,47	-73,41	468,70	475,40	2,960	2,665	-1,311
3 590,00	80,41	123,96	2 915,21	-78,94	476,85	484,11	2,960	2,667	-1,304
3 600,00	81,30	123,53	2 916,80	-84,42	485,06	492,88	2,960	2,668	-1,297
3 610,00	82,20	122,67	2 918,24	-89,83	493,35	501,72	3,731	2,707	-2,595
3 620,00	83,10	121,73	2 919,52	-95,11	501,74	510,65	3,883	2,719	-2,796
3 630,00	84,01	120,81	2 920,64	-100,27	510,23	519,67	3,883	2,724	-2,785
3 640,00	84,92	119,88	2 921,60	-105,30	518,82	528,76	3,883	2,729	-2,776
3 650,00	85,82	119,53	2 922,41	-110,23	527,48	537,92	2,899	2,700	-1,060
3 660,00	86,72	119,26	2 923,06	-115,13	536,18	547,11	2,812	2,695	-0,803
3 670,00	87,62	118,99	2 923,55	-119,99	544,90	556,32	2,812	2,696	-0,801
3 680,00	88,52	118,73	2 923,89	-124,82	553,66	565,55	2,812	2,696	-0,800
3 690,00	88,95	118,53	2 924,10	-129,60	562,43	574,81	1,419	1,288	-0,594

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
3 700,00	89,28	118,34	2 924,26	-134,37	571,22	584,08	1,142	1,000	-0,552
3 710,00	89,61	118,16	2 924,36	-139,10	580,03	593,36	1,142	1,000	-0,552
3 720,00	89,95	117,97	2 924,39	-143,80	588,86	602,65	1,142	1,000	-0,552
3 730,00	89,82	117,82	2 924,41	-148,48	597,69	611,96	0,599	-0,366	-0,474
3 740,00	89,60	117,67	2 924,46	-153,14	606,54	621,27	0,824	-0,687	-0,455
3 750,00	89,37	117,51	2 924,55	-157,77	615,41	630,59	0,824	-0,687	-0,455
3 760,00	89,14	117,36	2 924,68	-162,38	624,28	639,93	0,824	-0,687	-0,455
3 770,00	89,09	117,26	2 924,84	-166,96	633,17	649,27	0,342	-0,138	-0,313
3 780,00	89,09	117,17	2 924,99	-171,53	642,06	658,61	0,276	0,007	-0,275
3 790,00	89,10	117,07	2 925,15	-176,09	650,96	667,96	0,276	0,007	-0,275
3 800,00	89,10	116,98	2 925,31	-180,64	659,86	677,32	0,276	0,007	-0,275
3 810,00	88,98	117,13	2 925,47	-185,18	668,77	686,68	0,573	-0,349	0,454
3 820,00	88,83	117,36	2 925,67	-189,76	677,66	696,02	0,826	-0,462	0,685
3 830,00	88,68	117,59	2 925,88	-194,37	686,53	705,35	0,826	-0,462	0,685
3 840,00	88,52	117,82	2 926,13	-199,02	695,38	714,66	0,826	-0,461	0,685
3 850,00	87,68	116,86	2 926,44	-203,63	704,25	723,99	3,834	-2,534	-2,879
3 860,00	86,58	115,46	2 926,94	-208,04	713,21	733,39	5,338	-3,297	-4,203
3 870,00	85,48	114,05	2 927,64	-212,21	722,27	742,86	5,338	-3,291	-4,213
3 880,00	84,39	112,64	2 928,52	-216,16	731,41	752,38	5,338	-3,283	-4,226
3 890,00	83,75	111,62	2 929,57	-219,89	740,63	761,96	3,602	-1,918	-3,065
3 900,00	83,41	110,86	2 930,69	-223,50	749,89	771,57	2,486	-1,011	-2,286
3 910,00	83,07	110,10	2 931,86	-226,97	759,20	781,20	2,486	-1,007	-2,289
3 920,00	82,74	109,33	2 933,10	-230,32	768,54	790,85	2,486	-1,004	-2,292
3 930,00	82,24	108,45	2 934,40	-233,53	777,92	800,53	3,042	-1,508	-2,665
3 940,00	81,60	107,45	2 935,81	-236,58	787,34	810,23	3,515	-1,917	-2,975
3 950,00	80,96	106,46	2 937,32	-239,47	796,79	819,95	3,515	-1,910	-2,985
3 960,00	80,33	105,46	2 938,95	-242,18	806,28	829,68	3,515	-1,902	-2,996
3 970,00	79,73	104,54	2 940,68	-244,73	815,79	839,42	3,242	-1,781	-2,751
3 980,00	79,17	103,69	2 942,51	-247,12	825,32	849,16	3,029	-1,685	-2,560
3 990,00	78,61	102,83	2 944,44	-249,37	834,87	858,90	3,029	-1,678	-2,569
4 000,00	78,06	101,97	2 946,46	-251,48	844,44	868,64	3,029	-1,671	-2,579
4 010,00	77,43	101,23	2 948,58	-253,44	854,01	878,37	2,879	-1,866	-2,244
4 020,00	76,75	100,58	2 950,81	-255,28	863,58	888,09	2,783	-2,047	-1,934
4 030,00	76,07	99,93	2 953,16	-257,01	873,15	897,79	2,783	-2,043	-1,945
4 040,00	75,39	99,28	2 955,63	-258,63	882,70	907,47	2,783	-2,037	-1,956
4 050,00	75,19	98,81	2 958,18	-260,14	892,25	917,12	1,503	-0,604	-1,423
4 060,00	75,45	98,51	2 960,72	-261,60	901,82	926,79	1,162	0,769	-0,900
4 070,00	75,70	98,21	2 963,21	-263,01	911,40	936,47	1,162	0,771	-0,898
4 080,00	75,96	97,91	2 965,66	-264,37	921,00	946,16	1,162	0,772	-0,896
4 090,00	76,21	97,51	2 968,06	-265,67	930,62	955,86	1,371	0,745	-1,187
4 100,00	76,45	97,02	2 970,43	-266,90	940,26	965,58	1,605	0,719	-1,477
4 110,00	76,69	96,53	2 972,75	-268,05	949,91	975,30	1,605	0,722	-1,474
4 120,00	76,93	96,04	2 975,03	-269,11	959,59	985,04	1,605	0,725	-1,471

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
4 130,00	77,30	95,50	2 977,26	-270,10	969,29	994,79	1,928	1,105	-1,621
4 140,00	77,79	94,91	2 979,42	-270,98	979,01	1 004,55	2,263	1,470	-1,762
4 150,00	78,28	94,33	2 981,49	-271,77	988,76	1 014,33	2,263	1,474	-1,755
4 160,00	78,77	93,74	2 983,48	-272,46	998,54	1 024,12	2,263	1,478	-1,749
4 170,00	79,16	93,20	2 985,39	-273,05	1 008,34	1 033,92	1,987	1,160	-1,643
4 180,00	79,42	92,69	2 987,25	-273,56	1 018,15	1 043,73	1,689	0,787	-1,521
4 190,00	79,69	92,18	2 989,06	-273,98	1 027,98	1 053,54	1,689	0,789	-1,519
4 200,00	79,95	91,68	2 990,83	-274,31	1 037,81	1 063,35	1,689	0,791	-1,516
4 210,00	80,10	91,12	2 992,56	-274,55	1 047,66	1 073,17	1,712	0,446	-1,678
4 220,00	80,09	90,49	2 994,28	-274,69	1 057,51	1 082,97	1,867	-0,011	-1,895
4 230,00	80,09	89,85	2 996,00	-274,72	1 067,36	1 092,76	1,867	-0,008	-1,896
4 240,00	80,09	89,22	2 997,72	-274,64	1 077,21	1 102,54	1,867	-0,004	-1,896
4 250,00	79,98	88,56	2 999,44	-274,45	1 087,06	1 112,31	1,993	-0,341	-1,993
4 260,00	79,70	87,85	3 001,21	-274,14	1 096,90	1 122,05	2,260	-0,828	-2,137
4 270,00	79,43	87,13	3 003,02	-273,71	1 106,72	1 131,77	2,260	-0,823	-2,140
4 280,00	79,15	86,42	3 004,88	-273,16	1 116,53	1 141,46	2,260	-0,818	-2,144
4 290,00	78,86	85,55	3 006,78	-272,48	1 126,33	1 151,11	2,705	-0,892	-2,602
4 300,00	78,52	84,41	3 008,75	-271,62	1 136,09	1 160,72	3,501	-1,021	-3,416
4 310,00	78,18	83,27	3 010,77	-270,57	1 145,83	1 170,28	3,501	-1,008	-3,424
4 320,00	77,85	82,13	3 012,84	-269,33	1 155,53	1 179,79	3,501	-0,994	-3,432
4 330,00	77,44	81,12	3 014,98	-267,89	1 165,20	1 189,23	3,188	-1,215	-3,017
4 340,00	76,83	80,50	3 017,20	-266,34	1 174,82	1 198,62	2,599	-1,845	-1,879
4 350,00	76,22	79,87	3 019,53	-264,68	1 184,40	1 207,96	2,599	-1,840	-1,888
4 360,00	75,60	79,23	3 021,97	-262,92	1 193,94	1 217,24	2,599	-1,835	-1,898
4 370,00	74,88	78,69	3 024,51	-261,06	1 203,43	1 226,47	2,689	-2,170	-1,642
4 380,00	73,80	78,42	3 027,21	-259,15	1 212,87	1 235,63	3,336	-3,245	-0,806
4 390,00	72,72	78,15	3 030,09	-257,21	1 222,25	1 244,73	3,336	-3,244	-0,815
4 400,00	71,64	77,87	3 033,15	-255,23	1 231,56	1 253,77	3,336	-3,243	-0,825
4 410,00	70,57	77,58	3 036,39	-253,22	1 240,80	1 262,73	3,310	-3,205	-0,876
4 420,00	69,56	77,22	3 039,80	-251,17	1 249,98	1 271,62	3,194	-3,026	-1,090
4 430,00	68,55	76,85	3 043,37	-249,07	1 259,08	1 280,43	3,194	-3,023	-1,104
4 440,00	67,55	76,47	3 047,11	-246,93	1 268,10	1 289,16	3,194	-3,021	-1,120
4 450,00	66,50	76,12	3 051,01	-244,75	1 277,05	1 297,81	3,286	-3,139	-1,056
4 460,00	65,26	75,90	3 055,10	-242,54	1 285,90	1 306,36	3,780	-3,729	-0,678
4 470,00	64,01	75,67	3 059,38	-240,32	1 294,66	1 314,82	3,780	-3,728	-0,692
4 480,00	62,77	75,43	3 063,86	-238,09	1 303,32	1 323,17	3,780	-3,727	-0,707
4 490,00	61,55	75,15	3 068,53	-235,85	1 311,87	1 331,42	3,748	-3,675	-0,832
4 500,00	60,40	74,69	3 073,39	-233,57	1 320,32	1 339,56	3,634	-3,424	-1,392
4 510,00	59,26	74,21	3 078,41	-231,26	1 328,65	1 347,58	3,634	-3,419	-1,424
4 520,00	58,13	73,73	3 083,61	-228,90	1 336,86	1 355,48	3,634	-3,414	-1,458
4 530,00	57,00	73,24	3 088,97	-226,50	1 344,95	1 363,25	3,594	-3,377	-1,456
4 540,00	55,94	72,82	3 094,49	-224,07	1 352,92	1 370,90	3,351	-3,184	-1,254
4 550,00	54,88	72,40	3 100,17	-221,61	1 360,78	1 378,43	3,351	-3,179	-1,286

Statoil
Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	MI @ 54,90m (MI)
Site:	Volve F	MD Reference:	MI @ 54,90m (MI)
Well:	F-11	North Reference:	Grid
Wellbore:	15/9-F-11 B	Survey Calculation Method:	Minimum Curvature
Design:	15/9-F-11 B	Database:	Production EDM P246N

Survey

Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)
4 560,00	53,82	71,96	3 106,00	-219,12	1 368,51	1 385,85	3,351	-3,175	-1,320
4 570,00	52,78	71,50	3 111,98	-216,60	1 376,13	1 393,13	3,323	-3,137	-1,367
4 580,00	51,86	70,99	3 118,09	-214,06	1 383,62	1 400,29	3,012	-2,758	-1,531
4 590,00	50,94	70,47	3 124,33	-211,48	1 391,00	1 407,34	3,012	-2,751	-1,570
4 600,00	50,03	69,93	3 130,69	-208,87	1 398,25	1 414,26	3,012	-2,744	-1,611
4 610,00	49,10	69,38	3 137,18	-206,22	1 405,39	1 421,06	3,032	-2,764	-1,637
4 620,00	47,96	68,96	3 143,80	-203,56	1 412,40	1 427,72	3,559	-3,432	-1,261
4 630,00	46,82	68,53	3 150,57	-200,89	1 419,25	1 434,24	3,559	-3,427	-1,307
4 640,00	45,68	68,08	3 157,48	-198,22	1 425,97	1 440,61	3,559	-3,422	-1,357
4 650,00	44,54	67,61	3 164,54	-195,55	1 432,53	1 446,83	3,560	-3,417	-1,411
4 660,00	43,39	67,10	3 171,74	-192,88	1 438,93	1 452,90	3,611	-3,457	-1,507
4 670,00	42,24	66,58	3 179,07	-190,20	1 445,18	1 458,81	3,611	-3,450	-1,572
4 680,00	41,09	66,03	3 186,55	-187,53	1 451,27	1 464,57	3,611	-3,442	-1,643
4 690,00	39,94	65,46	3 194,15	-184,86	1 457,19	1 470,16	3,611	-3,434	-1,720
4 700,00	39,31	65,20	3 201,85	-182,20	1 462,99	1 475,62	1,958	-1,895	-0,772
4 710,00	38,70	64,95	3 209,62	-179,55	1 468,69	1 480,99	1,907	-1,846	-0,761
4 720,00	38,08	64,69	3 217,46	-176,91	1 474,31	1 486,28	1,907	-1,844	-0,782
4 730,00	37,47	64,42	3 225,36	-174,27	1 479,85	1 491,49	1,907	-1,842	-0,803
4 740,00	37,21	64,33	3 233,31	-171,65	1 485,31	1 496,63	0,779	-0,761	-0,272
4 750,00	36,97	64,24	3 241,29	-169,03	1 490,75	1 501,74	0,744	-0,728	-0,258
4 760,00	36,80	64,18	3 249,29	-166,42	1 496,15	1 506,81	0,528	-0,516	-0,185
4 770,00	36,80	64,18	3 257,30	-163,81	1 501,54	1 511,88	0,000	0,000	0,000

Casing Points

Measured Depth (m)	Vertical Depth (m)	Name	Casing Diameter (in)	Hole Diameter (in)
202,00	202,00	30"	30,000	36,000
1 357,70	1 334,20	20"	20,000	26,000
2 570,68	2 440,85	14"	14,000	17,500
3 192,40	2 780,54	9 5/8"x10 3/4"	9,625	12,250
4 768,00	3 255,70	7"	7,000	8,500

Design Annotations

Measured Depth (m)	Vertical Depth (m)	Local Coordinates +N/-S (m)	Local Coordinates +E/-W (m)	Comment
4 770,00	3 257,30	-163,81	1 501,54	Projection to TD

7 Appendix 2: Operational experience

Experience report written offshore

7.1 Appendix 2.1: Operational listing

WELLBORE ID: NO 15/9-F-11 B
INTERVAL: Wellbore preparation
START TIME: 16.Jun.2013 13:30
END TIME: 21.Jun.2013 07:00

Report date	Description
17.Jun.2013	Set liner packer and pressure tested same to 200 bar for 5 minutes - ok. Pulled out of hole with liner hanger running tool on 5 1/2" drill pipe from 3100m MD. Rigged up flush BHA and ran in hole to 167m MD. Troubleshot top drive brake release failure.
18.Jun.2013	Troubleshot brake release failure on top drive.
19.Jun.2013	Flushed BOP and riser. Pressure tested well to 270 bar. Ran in hole with cleanout assembly to 2834m MD.
20.Jun.2013	Ran in hole with cleanout assembly from 2834m MD to 4694m MD. Washed and polished PBR. Performed inflow test from 94 bar to 4 bar in steps of 30 bar. Displaced well to 1.03 sg brine.
21.Jun.2013	Pulled out of hole with cleanout assembly from 4150m MD to 2796m MD. Stopped due to damaged junction box on aft PRS. Pulled out of hole with cleanout assembly from 2796m MD to 1512m MD.

INTERVAL: Other
START TIME: 21.Jun.2013 21:00
END TIME: 28.Jun.2013 01:00

Report date	Description
22.Jun.2013	Pulled out of hole with cleanout string from 1512m MD. Displaced riser to seawater. Installed Safelock plug at 635m MD, and pressure tested same to 200 bar (from below) and 345 bar (from above). Retrieved 10 3/4" wearbushing. Installed lock down assembly. Nippled down diverter and BOP.
23.Jun.2013	Nippled down diverter and BOP. Installed claxton tool to NT-2 connector. Rigged down tensioning cylinders and tension ring. Installed guide wires.

Report date	Description
24.Jun.2013	Ran guide wires and attached same in wellhead template. Disconnected riser connector. Pulled out of hole with high pressure drilling riser.
25.Jun.2013	Pulled high pressure drilling riser and laid out to cantilever. Ran production riser.
26.Jun.2013	Ran production riser to wellhead. Landed and locked tieback connector on wellhead. Pressure tested production riser and tieback connector to 200 bar.
27.Jun.2013	Installed tie back connector. Installed riser centralizers, tension ring and tension cylinders. Cut off and bevelled riser tension joint. Landed multibowl.

INTERVAL: Tie-back
START TIME: 28.Jun.2013 01:00
END TIME: 30.Jun.2013 07:15

Report date	Description
28.Jun.2013	Replaced annular preventer. Installed surface riser and BOP.
29.Jun.2013	Retrieved annulus insert. RIH with tie-back string.
30.Jun.2013	Installed tie-back connector and pressure tested same. Installed adjustable surface hanger and tested same. Ran seal assembly.

INTERVAL: Other
START TIME: 29.Jun.2013 23:00
END TIME: 30.Jun.2013 14:15

Report date	Description
30.Jun.2013	Installed tie-back connector and pressure tested same. Installed adjustable surface hanger and tested same. Ran seal assembly.

INTERVAL: Run Completion
START TIME: 30.Jun.2013 14:15
END TIME: 04.Jul.2013 20:00

Report date	Description

Report date	Description
01.Jul.2013	Slipped and cut drill line. Pulled Safelock plug. Rigged up for completion.
02.Jul.2013	Rigged up to run completion. Ran completion.
03.Jul.2013	RIH with 7" tubing from 971 m to 2035 m
04.Jul.2013	RIH with 7" tubing from 2035 m to 2612 m. Picked up DHSV.

INTERVAL: Install Tubing Hanger
START TIME: 04.Jul.2013 20:00
END TIME: 06.Jul.2013 12:30

Report date	Description
05.Jul.2013	RIH with completion string to TOL. Pulled back and installed spaceout pups Installed tubing hanger.
06.Jul.2013	Landed and tested hanger. Sat production packer. Retrieved EVO plug. Pressure tested production packer from below.

INTERVAL: Install X-mas Tree
START TIME: 06.Jul.2013 12:30
END TIME: 09.Jul.2013 00:45

Report date	Description
07.Jul.2013	Ran BPV plug. Nippled down BOP. Prepared to install XMT.
08.Jul.2013	Terminated XMT. Performed planned maintenance .

INTERVAL: Perforation
START TIME: 09.Jul.2013 00:45
END TIME: 21.Jul.2013 23:00

Report date	Description
09.Jul.2013	Terminated and tested XMT. Rigged up for wireline.
10.Jul.2013	Rigged up for wireline. Prepared equipment for perforation. Made up preforation tool string.

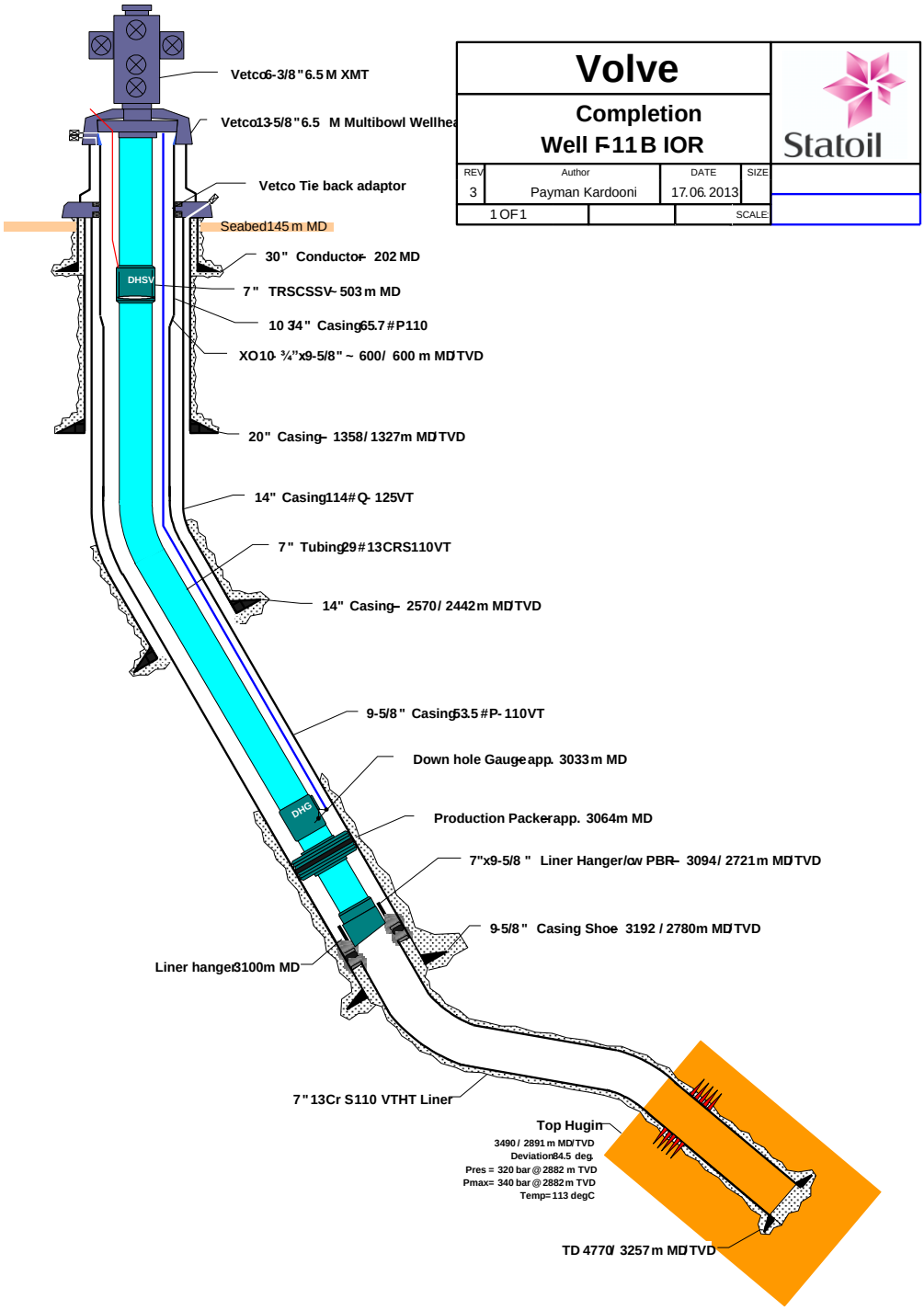
Report date	Description
11.Jul.2013	RIH with BHA # 1 from surface to 3700 m. Worked passed restriction at 3700 m and RIH to 3919 m. Lost communication with tractor. POOH and L/O BHA # 1. P/U and M/U BHA # 1RR and RIH to 3471 m.
12.Jul.2013	RIH with perforating BHA # 1RR to perforating depth at 4539 m. Perforated interval # 1. POOH to tool string inside tool catcher. Wireline cable released from cablehead. Broke lubricator and secured tool string with C-plate. Verified guns fired and L/D lubricator and BHA # 1RR.
13.Jul.2013	M/U perforation BHA # 2 and RIH to perforation depth. Perforated interval # 2 from 4480 m to 4500 m. POOH with perforation BHA # 2 and L/O guns.
14.Jul.2013	L/O perforation BHA # 2. P/U and M/U perforation BHA # 3. RIH with BHA # 3 to 4420 m. Unable to communicate with tools. POOH with BHA # 3. Troubleshoot communication failure. Build new cablehead and P/U perforation BHA # 3RR.
15.Jul.2013	RIH with perforation BHA # 3RR. Correlated and positioned guns between 4355 m to 4367 m. Perforated interval # 3 and POOH. L/O perforation BHA # 3RR and re-buil cablehead. P/U and M/U perforation BHA # 4 and RIH to 3825 m
16.Jul.2013	RIH with perforation BHA # 4 to 4470 m. Lost communication to tools. POOH with perforation BHA # 4 and L/O same. L/D lubricator and changed out wireline wench. Installed back-up wench and test wireline package, No Go. slipped and cut 800 m wireline caple and tested wireline. P/U and M/U perforation BHA # 4RR.
17.Jul.2013	P/U and M/U perforation BHA # 4RR1. RIH from surface to 3670 m. Worked assembly from 3670 m to 3921 m. Unable to pass 3921 m. POOH and L/O perforation BHA 4RR1. Changed out greasehead on wireline and RIH with perforation BHA # 4RR2 to 500 m.
18.Jul.2013	RIH with perforation BHA # 4RR2. Attempted to correlate and position gun assembly, No Go. POOH with perforation BHA # 4RR2 to surface and L/O same. Changed out VPM-g, PTA, cablehead and tractor. P/U and M/U perforation BHA # 4RR3. RIH and positioned gun assembly. Perforated interval # 4 from 4304 m to 4316 m and POOH to surface.
19.Jul.2013	L/O perforation BHA # 4RR3. P/U and M/U perforation BHA # 5. RIH and perforated interval # 5. POOH from 4295 m to surface with perforation BHA # 5 and L/O same.
20.Jul.2013	RIH with perforation BHA # 6 and positioned gun assembly at interval # 6 from 4268 m to 4280 m. Perforated interval # 6 and POOH. L/O perforation BHA # 6 and re-build cablehead. P/U and M/U perforation BHA # 7. RIH with perforation BHA # 7 to 3693 m.
21.Jul.2013	RIH with perforation BHA # 7. Postioned guns and perforated interval # 7. POOH and L/O perforation BHA # 7. P/U and RIH perforation BHA # 8. Positioned perforation BHA # 8 and perforated interval # 8. POOH to surface perforation BHA # 8.

INTERVAL: Other (Offline Activity)
START TIME: 22.Jul.2013 14:00
END TIME: 23.Jul.2013 20:00

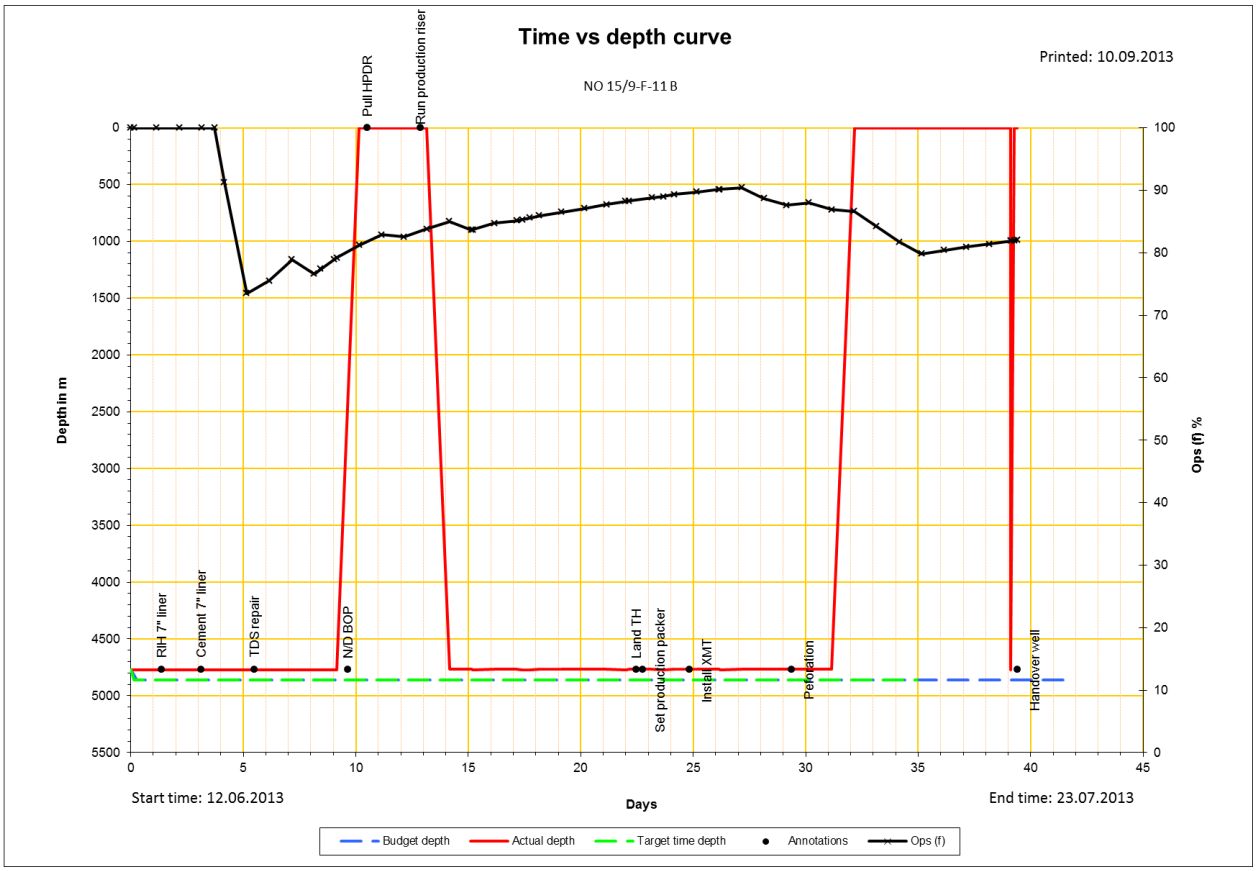
Report date	Description
23.Jul.2013	Performed pressure testing of valves on F-11 Xmas tree in preparation for handover to production department.

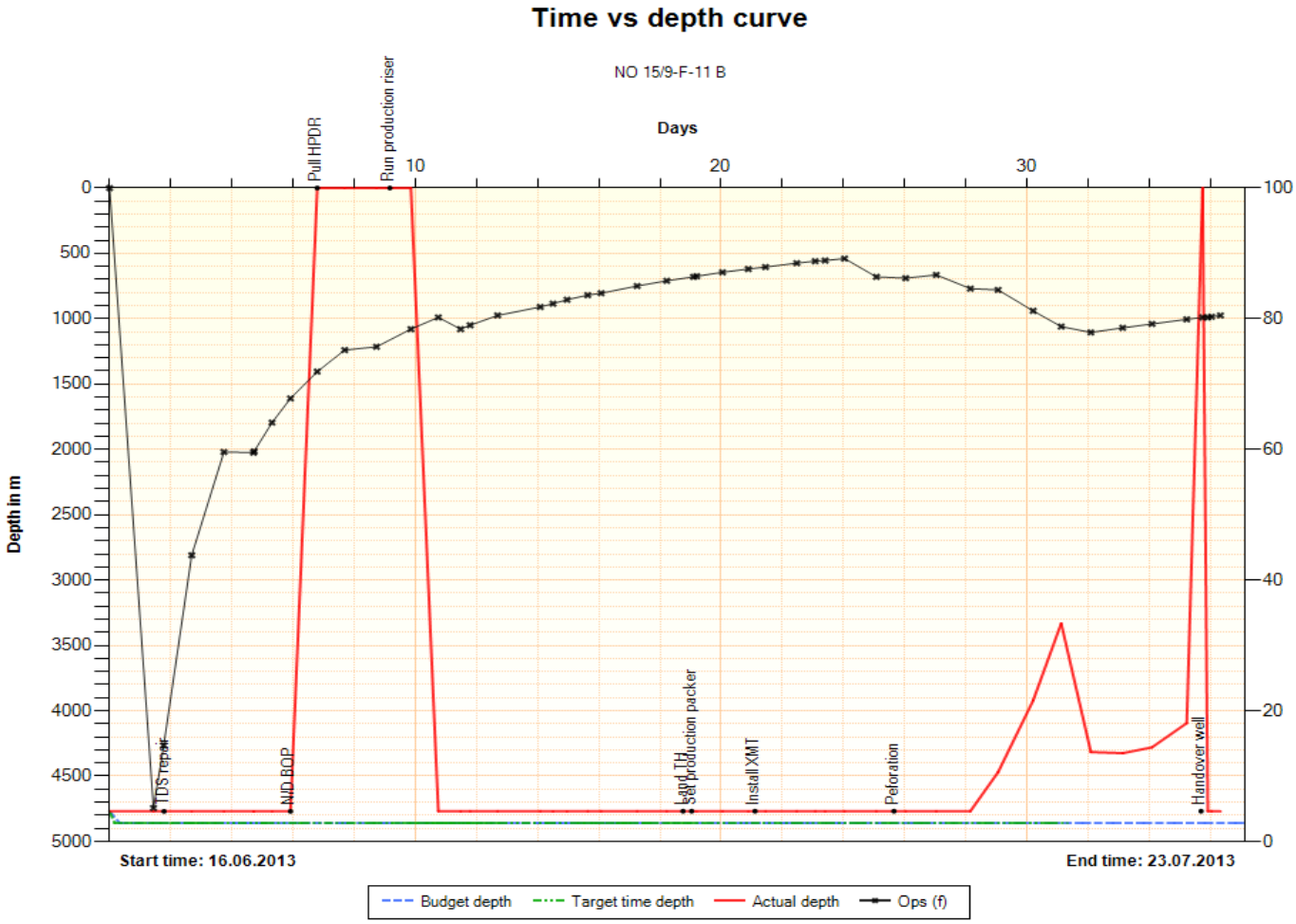
8 Appendix 3: Figures and tables

8.1 Wellbore schematic


Volve
Completion
Well F-11 B IOR
Statoil
REV 3
Author Payman Kardooni
DATE 17.06.2013
SIZE
1 OF 1
SCALE
Vetco 6-3/8" 6.5 M XMT
Vetco 13-5/8" 6.5 M Multibowl Wellhea
Vetco Tie back adaptor
Seabed 145 m MD
30" Conductor - 202 MD
DHSV
7" TRSCSSV - 503 m MD
10 3/4" Casing 65.7 # P110
XO 10-¾"x9-5/8" ~ 600/ 600 m MD/TVD
20" Casing - 1358/ 1327m MD/TVD
14" Casing 114# Q-125VT
7" Tubing 29 # 13CRS110VT
14" Casing - 2570/ 2442 m MD/TVD
9-5/8" Casing 53.5 # P-110VT
Down hole Gauge app. 3033 m MD
DHG
Production Packer app. 3064m MD
7"x9-5/8" Liner Hanger/ow PBR - 3094/ 2721 m MD/TVD
9-5/8" Casing Shoe 3192 / 2780m MD/TVD
Liner hanger 3100m MD
7" 13Cr S110 VTHT Liner
Top Hugin
3490 / 2891 m MD/TVD
Deviation 84.5 deg.
Pres = 320 bar @ 2882 m TVD
Pmax= 340 bar @ 2882 m TVD
Temp= 113 degC
TD 4770/ 3257 m MD/TVD

8.2 Time/depth curve




Time vs depth curve
NO 15/9-F-11 B
Days
Depth in m
Pull HPDR
Run production riser
TDS repair
N/D BOP
Land TH
Set production packer
Install XMT
Peforation
Handover well
Start time: 16.06.2013
End time: 23.07.2013
Budget depth
Target time depth
Actual depth
Ops (f)

8.3 Project planner

PROJECT NAME: NO 15/9 F-11B IOR
PROJECT NUMBER: T.O046V.DG.10068

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
16.Jun.2013 13:30	21.Jun.2013 07:00	99,0	4,1	59,0	2,5	77,0	113,7	4,7	45,8	Wellbore preparation [NO 15/9-F-11 B]	
16.Jun.2013 13:30	18.Jun.2013 08:15	8,0	0,3	4,0	0,2	5,0	42,8	1,8	36,3	36 Perform BOP/Riser/WH cleaning run.	Hall,Maersk,BHA
18.Jun.2013 08:15	18.Jun.2013 20:15	12,0	0,8	8,0	0,5	8,0	12,0	2,3	0,0	38 MU clean out assy for 7" liner and RIH on 3 1/2" DP.	Hall,Maersk,BHA
18.Jun.2013 08:15	18.Jun.2013 08:15		0,8		0,5	12,0	0,0	2,3	0,0	37 TDS repair	Hall,Maersk,BHA
18.Jun.2013 20:15	19.Jun.2013 03:30	14,0	1,4	8,0	0,8	10,0	7,3	2,6	0,0	39 MU 7" PBR tieback polish/topdress mill, well comm. tool and 9 5/8" casing cleanout assy and RIH on 5 1/2" DP.	Hall,Maersk
19.Jun.2013 03:30	19.Jun.2013 05:15	6,0	1,7	3,0	1,0	4,0	1,8	2,7	0,0	40 MU 10 3/4" casing clean out assy and RIH on 5 1/2" DP.	Hall,Maersk
19.Jun.2013 05:15	19.Jun.2013 18:30	14,0	2,3	8,0	1,3	4,0	13,3	3,2	0,0	41 Scrape 7"/9 5/8"/10 3/4" casing, dress/polish 7" liner top. Condition mud.	Hall,Maersk
19.Jun.2013 18:30	19.Jun.2013 21:30	6,0	2,5	3,0	1,4	6,0	3,0	3,3	0,0	42 Inflow test 7" liner with well commissioner and base oil in DP	Hall,Maersk
19.Jun.2013 21:30	20.Jun.2013 02:45	10,0	2,9	6,0	1,7	8,0	5,3	3,6	0,0	43 Pump wash pills and displace well to 1.03 sg NaCl until well is clean.	Hall,Maersk
20.Jun.2013 02:45	20.Jun.2013 03:15	12,0	3,4	8,0	2,0	6,0	0,5	3,6	0,0	44 Clean surface pits shakers and flowlines	Hall,Maersk
20.Jun.2013 03:15	20.Jun.2013 04:15	3,0	3,5	1,0	2,0	2,0	1,0	3,6	0,0	45 Leaktest well to 345 bar w/ 1,03 sg completion brine.	Hall,Maersk
20.Jun.2013 04:15	21.Jun.2013 07:00	14,0	4,1	10,0	2,5	12,0	26,7	4,7	9,5	46 POOH and LD BHA's. R/B 3 1/2" DP	Hall,Maersk,BHA
21.Jun.2013 07:00	21.Jun.2013 21:00	19,0	0,8	10,0	0,4	13,0	14,0	0,6	0,0	Other [NO 15/9-F-11 B]	
21.Jun.2013 07:00	21.Jun.2013 10:30	6,0	4,4	3,0	2,6	3,0	3,5	4,9	0,0	47 MU and RIH with 9 5/8" safelock plug.	Hall,Maersk

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
21.Jun.2013 10:30	21.Jun.2013 12:00	3,0	4,5	2,0	2,7	2,0	1,5	4,9	0,0	48 Set safelock plug at +/-620 m MDRT. Leak test from below to 345 bar.	Hall,Maersk
21.Jun.2013 12:00	21.Jun.2013 13:30	4,0	4,7	2,0	2,8	2,0	1,5	5,0	0,0	49 POOH and LD safelock setting tool.	Hall,Maersk
21.Jun.2013 13:30	21.Jun.2013 21:00	6,0	4,9	3,0	2,9	6,0	7,5	5,3	0,0	50 Pull wearbushing. Install 10 3/4" lock down sub.	Hall,Maersk
21.Jun.2013 21:00	28.Jun.2013 01:00	216,0	9,0	179,5	7,5	116,0	148,1	6,2	13,8	Other [NO 15/9-F-11 B]	
21.Jun.2013 21:00	22.Jun.2013 11:30	8,0	5,3	4,0	3,0	6,0	14,5	5,9	0,0	51 Nipple down Diverter and BOP.	Maersk
22.Jun.2013 11:30	22.Jun.2013 14:00	8,0	5,6	4,0	3,2	6,0	2,5	6,0	0,0	52 Run in and MU Claxton tool to NT-2 connector on High Pressure Drilling Riser	Maersk
22.Jun.2013 14:00	22.Jun.2013 14:00	36,0	7,1	36,0	4,7	0,0	0,0	6,0	0,0	53 WOW for pulling HPDR	Maersk
22.Jun.2013 14:00	23.Jun.2013 03:30	8,0	7,4	6,0	5,0	7,0	13,5	6,6	0,0	54 RD sylinders, tensioner ring and centralizer	Maersk
23.Jun.2013 03:30	23.Jun.2013 04:30	3,0	7,5	1,5	5,0	2,0	1,0	6,6	0,0	55 Disconnect HPDR from subsea wellhead. HEAVY LIFT OPERATION - NO PRODUCTION.	Maersk
23.Jun.2013 04:30	23.Jun.2013 08:30	4,0	7,7	2,0	5,1	2,0	4,0	6,8	0,0	56 R/U for pulling HPDR	Maersk
23.Jun.2013 08:30	24.Jun.2013 05:30	26,0	8,8	26,0	6,2	18,0	21,0	7,7	0,0	57 POOH with HPDR. HEAVY LIFT OPERATION - NO PRODUCTION.	Maersk
24.Jun.2013 05:30	24.Jun.2013 08:45	4,0	9,0	2,0	6,3	3,0	3,3	7,8	0,0	58 Prepare for running production riser	Maersk
24.Jun.2013 08:45	24.Jun.2013 08:45	32,0	10,3	36,0	7,8	0,0	0,0	7,8	0,0	59 WOW for running production riser	Maersk
24.Jun.2013 08:45	25.Jun.2013 17:30	24,0	11,3	16,0	8,4	18,0	32,7	9,2	5,3	60 Run production riser. HEAVY LIFT OPERATION - NO PRODUCTION	Maersk
25.Jun.2013 17:30	26.Jun.2013 01:15	7,0	11,6	5,0	8,6	6,0	7,8	9,5	0,0	61 Land the riser and lock the TBC	Maersk
26.Jun.2013 01:15	26.Jun.2013 10:00	18,0	12,3	14,0	9,2	16,0	8,8	9,9	0,0	62 Install Centralisers, boat collision tool, Tensioner ring assy and tension sylinders.	Maersk
26.Jun.2013 10:00	26.Jun.2013 15:30	12,0	12,8	8,0	9,6	10,0	5,5	10,1	0,0	63 Cut (and bewel) the riser tension joint	Maersk
26.Jun.2013 15:30	27.Jun.2013 07:30	8,0	13,2	6,0	9,8	7,0	16,0	10,8	0,0	64 Install surface wellhead (multibowl). Test BT seals.	Maersk
27.Jun.2013 07:30	27.Jun.2013 18:00	8,0	13,5	6,0	10,1	7,0	10,5	11,2	8,5	65 Install surface riser	Maersk
27.Jun.2013 18:00	28.Jun.2013 01:00	10,0	13,9	7,0	10,4	8,0	7,0	11,5	0,0	66 N/U BOP and diverter	Maersk
28.Jun.2013 01:00	30.Jun.2013 07:15	55,5	2,3	29,0	1,2	40,0	59,2	2,5	2,0	Tie-back [NO 15/9-F-11 B]	
28.Jun.2013 01:00	28.Jun.2013 02:45	2,0	14,0	1,0	10,4	1,0	1,8	11,6	0,0	67 Install Bowl protector in surface wellhead	Hall,Vetco,Maersk
28.Jun.2013 02:45	28.Jun.2013 02:45	2,0	14,1	1,0	10,4	0,0	0,0	11,8	0,0	68 Wash multibowl.	Hall,Vetco,Maersk
28.Jun.2013 02:45	28.Jun.2013 08:30	6,0	14,3	4,0	10,6	5,0	5,8	11,8	0,0	69 RIH and retrieve Annulus insert from subsea wellhead.	Hall,Vetco,Maersk

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
28.Jun.2013 08:30	28.Jun.2013 13:45	3,0	14,5	2,0	10,7	2,0	5,3	12,0	0,0	70 RIH and clean subsea wellhead and displace to 1.03sg inhibited brine	Hall,Vetco,Maersk
28.Jun.2013 13:45	28.Jun.2013 14:30	2,0	14,5	1,0	10,7	1,0	0,8	12,0	0,0	71 RIH and Retrieve Bowl protector from surface wellhead	Hall,Vetco,Maersk
28.Jun.2013 14:30	28.Jun.2013 17:15	6,0	14,8	2,0	10,8	2,0	2,8	12,2	0,0	72 R/U to run 10 3/4" tie back string	Hall,Vetco,Maersk,Weath
28.Jun.2013 17:15	28.Jun.2013 20:00	0,0	14,8		10,8	0,5	2,8	12,3	0,0	73 M/U and Rack back DQ adjustable setting tool	Hall,Vetco,Maersk
28.Jun.2013 20:00	28.Jun.2013 23:15	4,0	15,0	2,0	10,9	3,0	3,3	12,4	0,0	74 RIH with subsea tieback connector on 10 3/4" casing	Hall,Vetco,Maersk,Weath
28.Jun.2013 23:15	29.Jun.2013 00:00	3,0	15,1	2,0	11,0	0,5	0,8	12,4	0,0	75 P/U and M/U 10 3/4" surface hanger with DQ premade adjustment sub.	Hall,Vetco,Maersk,Weath
29.Jun.2013 00:00	29.Jun.2013 01:00	3,0	15,2	1,0	11,0	1,0	1,0	12,5	0,0	76 RIH with 10 3/4" tie back assembly and land off in subsea casing hanger	Hall,Vetco,Maersk,Weath
29.Jun.2013 01:00	29.Jun.2013 03:00	5,0	15,4	3,0	11,1	2,0	2,0	12,6	0,0	77 Discoonect casing hanger running tool and lay out landing string	Hall,Vetco,Maersk,Weath
29.Jun.2013 03:00	29.Jun.2013 03:30	3,0	15,5	1,0	11,2	0,5	0,5	12,6	0,0	78 P/U and M/U tie back adapter torque tool	Hall,Vetco,Maersk,Weath
29.Jun.2013 03:30	29.Jun.2013 05:30	5,0	15,7	2,0	11,3	2,0	2,0	12,7	0,0	79 RIH with torque tool and make up tie back connection	Hall,Vetco,Maersk,Weath
29.Jun.2013 05:30	29.Jun.2013 06:00	0,0	15,7		11,3	1,0	0,5	12,7	0,0	80 POOH with torque tool & lay out	Hall,Vetco,Maersk,Weath
29.Jun.2013 06:00	29.Jun.2013 08:45	0,0	15,7		11,3	2,0	2,8	12,8	0,0	81 P/U casing hanger running tool and RIH and M/U according to GE	Hall,Vetco,Maersk,Weath
29.Jun.2013 08:45	29.Jun.2013 10:45	0,0	15,7		11,3	1,5	2,0	12,9	0,0	82 Install 10-3/4" casing swedge and test tie back connection and adajustable sub inner seal	Hall,Vetco,Maersk,Weath
29.Jun.2013 10:45	29.Jun.2013 11:30	0,0	15,7		11,3	1,0	0,8	12,9	0,0	83 M/U DQ adjustable sub setting tool	Hall,Vetco,Maersk,Weath
29.Jun.2013 11:30	29.Jun.2013 12:00	0,0	15,7		11,3	1,0	0,5	12,9	0,0	84 RIH with DQ adjustable sub setting tool and set tension in tie back string	Hall,Vetco,Maersk,Weath
29.Jun.2013 12:00	29.Jun.2013 18:00	0,0	15,7		11,3	0,5	6,0	13,2	0,0	85 POOH with setting tool	Hall,Vetco,Maersk,Weath
29.Jun.2013 18:00	29.Jun.2013 20:15	0,0	15,7		11,3	1,0	2,3	13,3	0,0	86 P/U and M/U MS-T seat running tool	Hall,Vetco,Maersk,Weath
29.Jun.2013 20:15	30.Jun.2013 01:15	0,0	15,7		11,3	4,0	5,0	13,5	2,0	87 RIH with MS-T seal, set & test (externally) POOH with running tool and lay out	Hall,Vetco,Maersk,Weath
30.Jun.2013 01:15	30.Jun.2013 02:00	0,0	15,8		11,3	3,0	0,8	13,9	0,0	88 Close shear rams and pressure test BOP connection, adjustment sub inner seal & MS-t from above	Hall,Vetco,Maersk,Weath

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
30.Jun.2013 02:00	30.Jun.2013 03:00	6,0	16,1	3,0	11,4	2,0	1,0	13,9	0,0	89 Run secodary bowl protector into multibowl wellhead	Hall,Maersk,Weath
30.Jun.2013 03:00	30.Jun.2013 03:15	4,0	16,2	3,0	11,6	0,5	0,3	13,9	0,0	90 POOH and lay out bowl protector RT	Hall,Maersk,Weath
30.Jun.2013 07:15	30.Jun.2013 14:15	15,0	0,6	8,0	0,3	9,5	7,0	0,3	0,0	Other [NO 15/9-F-11 B]	
30.Jun.2013 07:15	30.Jun.2013 07:15	6,0	16,5	4,0	11,7	4,0	0,0	13,9	0,0	92 M/U and RIH with DP and VMB retrieving tool.	Maersk,Weath
30.Jun.2013 07:15	30.Jun.2013 10:45	3,0	16,6	1,0	11,8	1,5	3,5	14,1	0,0	93 Circulate out Hi-vis pill with packer fluid.	Hall,Maersk,Weath
30.Jun.2013 10:45	30.Jun.2013 14:15	6,0	16,9	3,0	11,9	4,0	3,5	14,2	0,0	94 Release VMB plug. POOH and L/D.	Hall,Maersk,Weath
30.Jun.2013 14:15	04.Jul.2013 20:00	100,0	4,2	96,0	4,0	66,0	101,9	4,2	0,8	Run Completion [NO 15/9-F-11 B]	
30.Jun.2013 14:15	30.Jun.2013 15:45	4,0	17,0	3,0	12,0	3,0	1,5	14,3	0,0	95 Pull bowl protector and wash surface wellhead and BOP	Hall,Maersk,Weath
30.Jun.2013 15:45	30.Jun.2013 19:00	4,0	17,2	3,0	12,1	3,0	3,3	14,4	0,0	96 Test BOP	Hall,Maersk
30.Jun.2013 19:00	01.Jul.2013 01:30	6,0	17,4	4,0	12,3	4,0	6,5	14,7	0,0	97 Prepare for running of completion string	Hall,Maersk
01.Jul.2013 01:30	01.Jul.2013 04:00	6,0	17,7	4,0	12,5	2,0	2,5	14,8	0,0	98 MU and RIH w/ completion assemblies (self aligning muleshoe, jnt w/7" EVO plug, production packer)	Hall,Maersk
01.Jul.2013 04:00	01.Jul.2013 05:45	8,0	18,0	6,0	12,7	6,0	1,8	14,9	0,0	99 MU DHPG. Make connections and test	Hall,Maersk
01.Jul.2013 05:45	01.Jul.2013 05:45	12,0	18,5	24,0	13,7	0,0	0,0	17,1	0,0	100 Completion technical allowance	Hall,Maersk
01.Jul.2013 05:45	03.Jul.2013 11:15	40,0	20,2	38,0	15,3	34,0	53,5	17,1	0,8	101 RIH with tubing to DHSV	Hall,Maersk
03.Jul.2013 11:15	03.Jul.2013 19:00	8,0	20,5	5,0	15,5	4,0	7,8	17,4	0,0	102 MU DHSV assy.	Maersk
03.Jul.2013 19:00	04.Jul.2013 14:30	6,0	20,8	6,0	15,8	6,0	19,5	18,2	0,0	103 RIH with tubing to TOL	Maersk
04.Jul.2013 14:30	04.Jul.2013 20:00	6,0	21,0	3,0	15,9	4,0	5,5	18,4	0,0	104 Pull back and install spaceout pups	Hall,Maersk
04.Jul.2013 20:00	06.Jul.2013 12:30	81,0	3,4	55,0	2,3	52,5	40,9	1,7	0,0	Install Tubing Hanger [NO 15/9-F-11 B]	
04.Jul.2013 20:00	05.Jul.2013 03:30	12,0	21,5	8,0	16,2	8,0	7,5	18,8	0,0	105 M/U TH and terminate 2 off lines for DHSV and 1 line for DHPG cable.	Hall,Maersk
05.Jul.2013 03:30	05.Jul.2013 07:30	10,0	21,9	8,0	16,6	8,0	4,0	18,9	0,0	106 M/U Comm. Collar and TH running tool. Land/test TH.	Hall,Maersk
05.Jul.2013 07:30	05.Jul.2013 10:00	3,0	22,1	1,0	16,6	2,0	2,5	19,0	0,0	107 Lowpressure tubing test and function test of DHSV	Hall,Maersk
05.Jul.2013 10:00	05.Jul.2013 10:15	2,0	22,1	1,0	16,6	0,5	0,3	19,0	0,0	108 Install 2" drop ball in tubing for setting packer	Hall,Maersk
05.Jul.2013 10:15	05.Jul.2013 10:45	4,0	22,3	3,0	16,8	1,0	0,5	19,1	0,0	109 Start circ. with low rate and circ. ball down to seat in EVO plug	Hall,Maersk
05.Jul.2013 10:45	05.Jul.2013 14:15	4,0	22,5	1,0	16,8	2,0	3,5	19,2	0,0	110 Set Production Packer and test tubing	Hall,Maersk

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
05.Jul.2013 14:15	05.Jul.2013 15:30	4,0	22,6	2,0	16,9	2,0	1,3	19,3	0,0	111 Leak test production packer from above and TH from below to 345 bar (A-annulus leak test)	Hall,Maersk
05.Jul.2013 15:30	05.Jul.2013 16:45	6,0	22,9	3,0	17,0	3,0	1,3	19,3	0,0	112 R/U W/L for pulling 7" EVO plug	Maersk,AWS
05.Jul.2013 16:45	05.Jul.2013 22:30	10,0	23,3	10,0	17,4	8,0	5,8	19,6	0,0	113 RIH with W/L and retrieve EVO plug, R/D W/L	Maersk,AWS
05.Jul.2013 22:30	06.Jul.2013 00:45	3,0	23,4	1,0	17,5	2,0	2,3	19,6	0,0	114 Inflowtest of DHSV	Maersk,AWS
06.Jul.2013 00:45	06.Jul.2013 04:30	3,0	23,6	2,0	17,6	4,0	3,8	19,8	0,0	116 POOH with Tbg Hgr running tool and landingstring	Maersk
06.Jul.2013 00:45	06.Jul.2013 00:45	3,0	23,7	1,0	17,6	0,0	0,0	19,6	0,0	115 Remove communication collar.	Maersk,AWS
06.Jul.2013 04:30	06.Jul.2013 06:15	3,0	23,8	2,0	17,7	2,0	1,8	19,9	0,0	117 Run BPV plug and set in tubing hanger	Maersk
06.Jul.2013 06:15	06.Jul.2013 12:30	14,0	24,4	12,0	18,2	10,0	6,3	20,1	0,0	118 Nipple down diverter, BOP and surface riser	Maersk
06.Jul.2013 12:30	09.Jul.2013 00:45	100,0	4,2	89,0	3,7	78,0	60,3	2,5	0,0	Install X-mas Tree [NO 15/9-F-11 B]	
06.Jul.2013 12:30	06.Jul.2013 21:30	12,0	24,9	5,0	18,4	8,0	9,0	20,5	0,0	120 Install dart in BPV, terminate control lines and install seals for X-mas tree.	Maersk
06.Jul.2013 12:30	06.Jul.2013 12:30	6,0	25,1	2,0	18,5	0,0	0,0	20,1	0,0	119 Lift down coflex production lines to XMT deck	Maersk
06.Jul.2013 21:30	07.Jul.2013 16:00	10,0	25,6	3,0	18,6	6,0	18,5	21,3	0,0	121 Install XMT and test same.	Maersk
07.Jul.2013 16:00	07.Jul.2013 16:00		25,6		18,6	0,0	0,0	21,3	0,0	122 Remove scaffolding	Maersk
07.Jul.2013 16:00	08.Jul.2013 16:30		25,6		18,6	50,0	24,5	22,3	0,0	123 *** Handover well to Maersk Production - Xmas tree work - receive well from Maersk ***	
08.Jul.2013 16:30	08.Jul.2013 16:30	6,0	25,8	3,0	18,7	0,0	0,0	22,3	0,0	124 Install Spacer Spool.	Maersk
08.Jul.2013 16:30	09.Jul.2013 00:45	6,0	26,1	2,0	18,8	4,0	8,3	22,6	0,0	126 Inflow test LMV and HMV. Remove tree cap. Retrieve dart and BPV. Reinstall and test tree cap..	Maersk
08.Jul.2013 16:30	08.Jul.2013 16:30	10,0	26,5	4,0	19,0	0,0	0,0	22,3	0,0	125 Install Flow Hose and test same.	Maersk
09.Jul.2013 00:45	21.Jul.2013 23:00	206,0	8,6	226,0	9,4	175,0	310,5	12,9	108,1	Perforation [NO 15/9-F-11 B]	
09.Jul.2013 00:45	09.Jul.2013 00:45	50,0	28,6	70,0	21,9	10,0	0,0	22,6	0,0	127 Commision XMT	Maersk
09.Jul.2013 00:45	09.Jul.2013 22:00	6,0	28,8	12,0	22,4	5,0	21,3	23,5	0,0	128 Rig up WL equipment	Schlum,Maersk,AWS
09.Jul.2013 22:00	12.Jul.2013 05:00	24,0	29,8	25,0	23,4	20,0	55,0	25,8	24,3	129 Perforate well Run #1. 4527-4539 (MD RKB)	Schlum,Maersk,AWS
12.Jul.2013 05:00	13.Jul.2013 01:30	24,0	30,8	25,0	24,5	20,0	20,5	26,7	0,0	130 Perforate well Run #2. 4488-4500 (MD RKB)	Schlum,Maersk,AWS
13.Jul.2013 01:30	14.Jul.2013 15:30	24,0	31,8	25,0	25,5	20,0	38,0	28,3	17,3	131 Perforate well Run #3. 4355-4367 (MD RKB)	Schlum,Maersk,AWS
14.Jul.2013 15:30	18.Jul.2013 05:00	24,0	32,8	25,0	26,6	20,0	85,5	31,8	66,5	132 Perforate well Run #4. 4304-4316 (MD RKB)	Schlum,Maersk,AWS
18.Jul.2013 05:00	18.Jul.2013 23:15	24,0	33,8	25,0	27,6	20,0	18,3	32,6	0,0	133 Perforate well Run #5. 4286-4295 (MD RKB)	Schlum,Maersk,AWS
18.Jul.2013 23:15	19.Jul.2013 16:15	24,0	34,8	25,0	28,6	20,0	17,0	33,3	0,0	134 Perforate well Run #6. 4268-4280 (MD RKB)	Schlum,Maersk,AWS

Start time	End time	Budget time hrs	Acc budget days	Target time hrs	Acc tech days	Plan time hrs	Act time hrs	Acc actual days	Down time	Description	Companies
19.Jul.2013 16:15	20.Jul.2013 10:30	24,0	35,8	25,0	29,7	20,0	18,3	34,1	0,0	135 Perforate well Run #7. 4063-4075 (MD RKB)	Schlum,Maersk,AWS
20.Jul.2013 10:30	21.Jul.2013 04:15	24,0	36,8	25,0	30,7	20,0	17,8	34,8	0,0	136 Perforate well Run #8. 4031-4043 (MD RKB)	Schlum,Maersk,AWS
21.Jul.2013 04:15	21.Jul.2013 04:15		36,8		30,7	0,0	0,0	34,8	0,0	137 Perforate well Run #9.	Schlum,Maersk,AWS
21.Jul.2013 04:15	21.Jul.2013 04:15		36,8		30,7	0,0	0,0	35,6	0,0	139 Perforate well Run #11.	
21.Jul.2013 04:15	21.Jul.2013 04:15		36,8		30,7	0,0	0,0	35,6	0,0	140 Perforate well Run #12.	
21.Jul.2013 04:15	21.Jul.2013 23:00	8,0	37,1	14,0	31,3	10,0	18,8	35,6	0,0	141 RD WL equipment.	Maersk,AWS

8.4 Borehole stability figure with MW and final pressure/fracture gradients

Updated figures in drilling program.

8.5 Bit record

No records found.
N/A

8.6 Bha list

Perforation Run no. 1

String component	OD in	ID in	Length m	Acc length m
4.5" Bottom Nose	4,5		0,11	0,11
4.5" Orienting weight OCD	4,5		3,3	3,41
4.5" Alignment Adapter	4,5		0,2	3,61
4.5" Alignment Adapter	4,5		0,11	3,72
4.5" 20' HSD OX Phasing 10-350 degFully Loaded PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF	4,5		6,34	10,06
4.5" Alignment Adapter	4,5		0,2	10,26
4.5" Alignment Adapter	4,5		0,11	10,37
4.5" 20' HSD OX Phasing 10-350 degFully Loaded PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF	4,5		6,34	16,71
4.5" Alignment Adapter	4,5		0,2	16,91
4.5" OX-Swivel	4,5		0,68	17,59
3 3/8" to 4.5" X-over	3,375		0,22	17,81
3-3/8 Long spacer housingASFS	3,375		0,4	18,21
3-3/8" Single Port Firing	3,375		0,14	18,35
ATPS-A ATPS Housing	3,375		0,61	18,96
WPSA-AA Shock Absorber	3,375		1,07	20,03
AH-83	3,375		0,16	20,19
PGGT-D (Powered Gun Gamma Ray Tool) GR/CCL	3,375		1,78	21,97
AH66	3,625		0,26	22,23
Short PEK-AH-591 3/8" 12 UNF Box to 3 3/8"	3,375		0,32	22,55
XO GO to 1 3/8" 12 UNF pin	1,688		0,07	22,62
VPMG	2,125		0,71	23,33
PowerTrac Advance 3182 sonde sections	3,35		4,59	27,92
RSS	3,125		1,3	29,22
AKWS Swivel	2,125		0,26	29,48
AKWS Head	1,438		0,38	29,86

Perforation Run no. 2-4

String component	OD in	ID in	Length m	Acc length m
4.5" Bottom Nose	4,5		0,11	0,11
4.5" Orienting weight OCD	4,5		3,3	3,41
4.5" Alignment Adapter	4,5		0,2	3,61
4.5" Alignment Adapter	4,5		0,11	3,72
4.5" 20' HSD OX Phasing 10-350 degFully Loaded PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF	4,5		6,34	10,06
4.5" Alignment Adapter	4,5		0,2	10,26
4.5" Alignment Adapter	4,5		0,11	10,37
4.5" 20' HSD OX Phasing 10-350 degFully Loaded PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF	4,5		6,34	16,71
4.5" Alignment Adapter	4,5		0,2	16,91
4.5" OX-Swivel	4,5		0,68	17,59
3 3/8" to 4.5" X-over	3,375		0,22	17,81
3-3/8 Long spacer housingASFS	3,375		0,4	18,21
3-3/8" Single Port Firing Head	3,375		0,14	18,35
ATPS-AATPS Housing	3,375		0,61	18,96
WPSA-AA Shock Absorber	3,375		1,07	20,03
AH-83	3,375		0,16	20,19
PGGT-D (Powered Gun Gamma Ray Tool) GR/CCL	3,375		1,78	21,97
AH66	3,625		0,26	22,23
Short PEK-AH-591 3/8" 12 UNF Box to 3 3/8"	3,375		0,32	22,55
XO GO to 1 3/8" 12 UNF pin	1,688		0,07	22,62
VPMG	2,125		0,71	23,33
PowerTrac Advance 3183 sonde sections	3,35		5,58	28,91
RSS	3,125		1,3	30,21
AKWS Swivel	2,125		0,26	30,47
AKWS Head	1,438		0,38	30,85

Perforation Run no. 5-8

String component	OD in	ID in	Length m	Acc length m
4.5" Bottom Nose	4,5		0,11	0,11
4.5" Orienting weightOCD	4,5		3,3	3,41
4.5" Alignment Adapter	4,5		0,2	3,61
4.5" Alignment Adapter	4,5		0,11	3,72
4.5" 30' HSD OX Phasing 10-350 degFully Loaded PowerJet Plus Low Debris 4504 HMXPhasing: 10-350 deg4 SPF	4,5		9,15	12,87
4.5" Alignment Adapter	4,5		0,2	13,07
4.5" OX-Swivel	4,5		0,68	13,75

String component	OD in	ID in	Length m	Acc length m
3 3/8" to 4.5" X-over	3,375		0,22	13,97
3-3/8 Long spacer housingASFS	3,375		0,4	14,37
3-3/8" Single Port Firing Head	3,375		0,14	14,51
ATPS-AATPS Housing	3,375		0,61	15,12
WPSA-AA Shock Absorber	3,375		1,07	16,19
AH-83	3,375		0,16	16,35
PGGT-D (Powered Gun Gamma Tool)GR/CCL	3,375		1,78	18,13
AH66	3,625		0,26	18,39
Short PEK-AH-591 3/8" 12 UNF Box to 3 3/8"	3,375		0,32	18,71
XO GO to 1 3/8" 12 UNF pin	1,688		0,07	18,78
VPMG	2,125		0,71	19,49
PowerTrac Advance 3183 sonde sections	3,35		5,58	25,07
RSS	3,125		1,3	26,37
AKWS Swivel	2,125		0,26	26,63
AKWS Head	1,438		0,38	27,01
PSTE-S				
SUB				

8.7 Drilling fluids

N/A

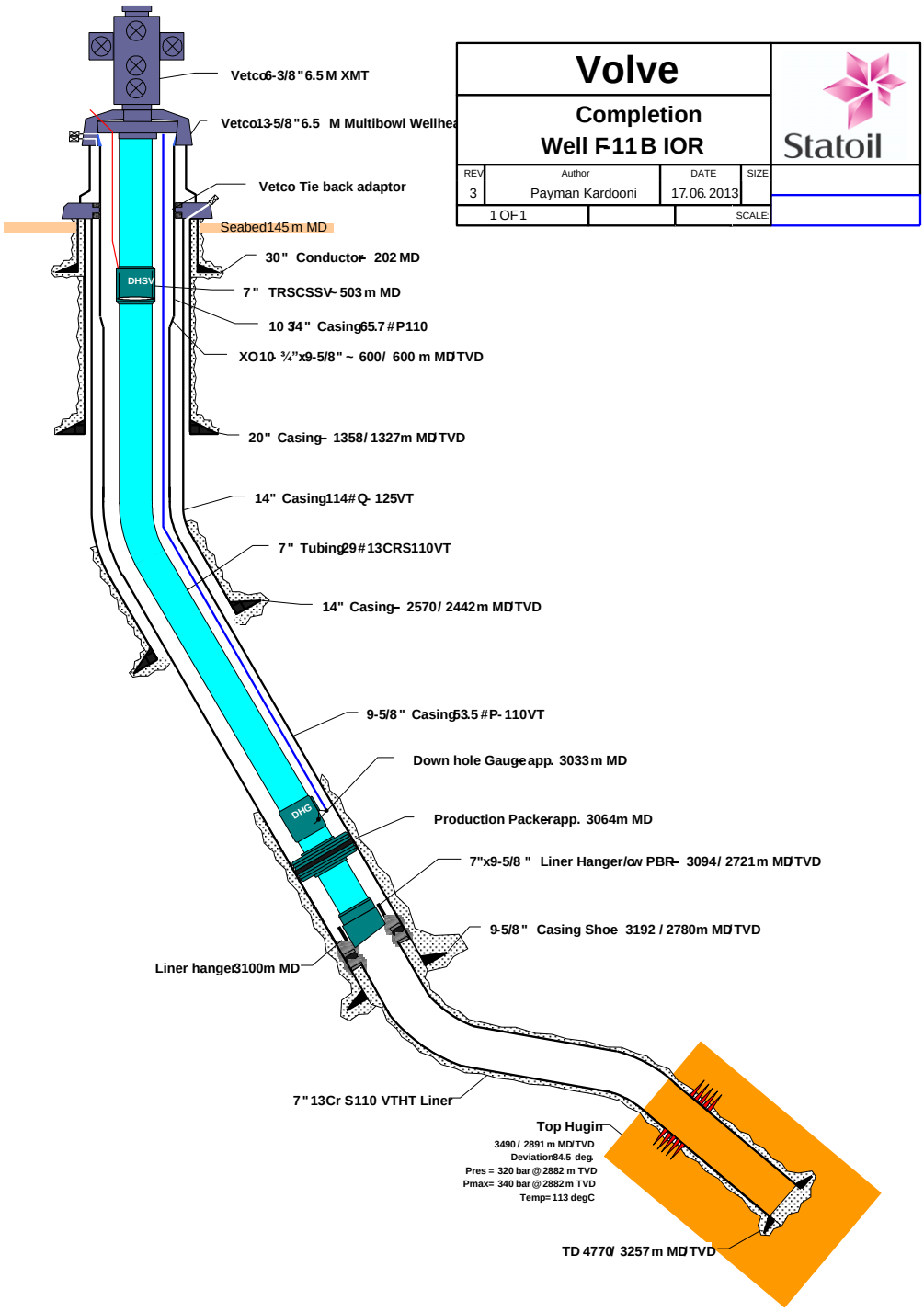
8.8 Cementing operations

N/A

8.9 Well schematic

N/A

8.10 Completion schematic



VOLVE

NO 15/9-F-11 B (NO 15/9-F-11 B IOR)

Compl.: 1.00

Rig: Maersk Inspirer **Status:** Oil Producer

Plan

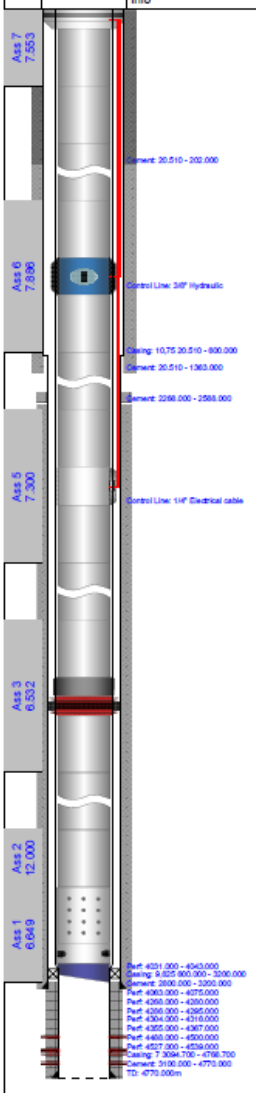
Completion F-11 B IOR

Page 1 of 2

Prep. by: PAYKA **Date:** 06.07.2013

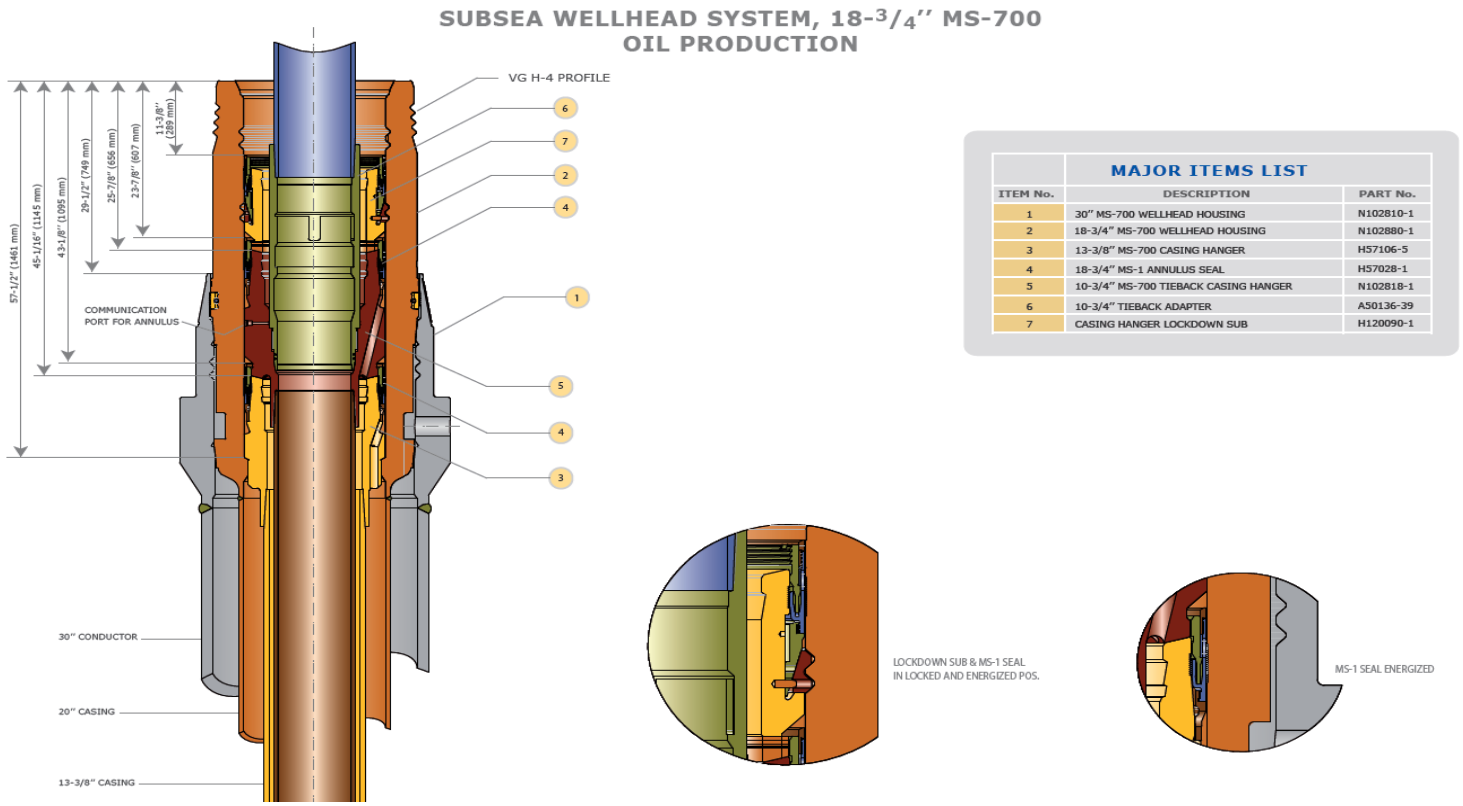
Rev.: 1.03

Last update: 27.08.2013 16:00:59



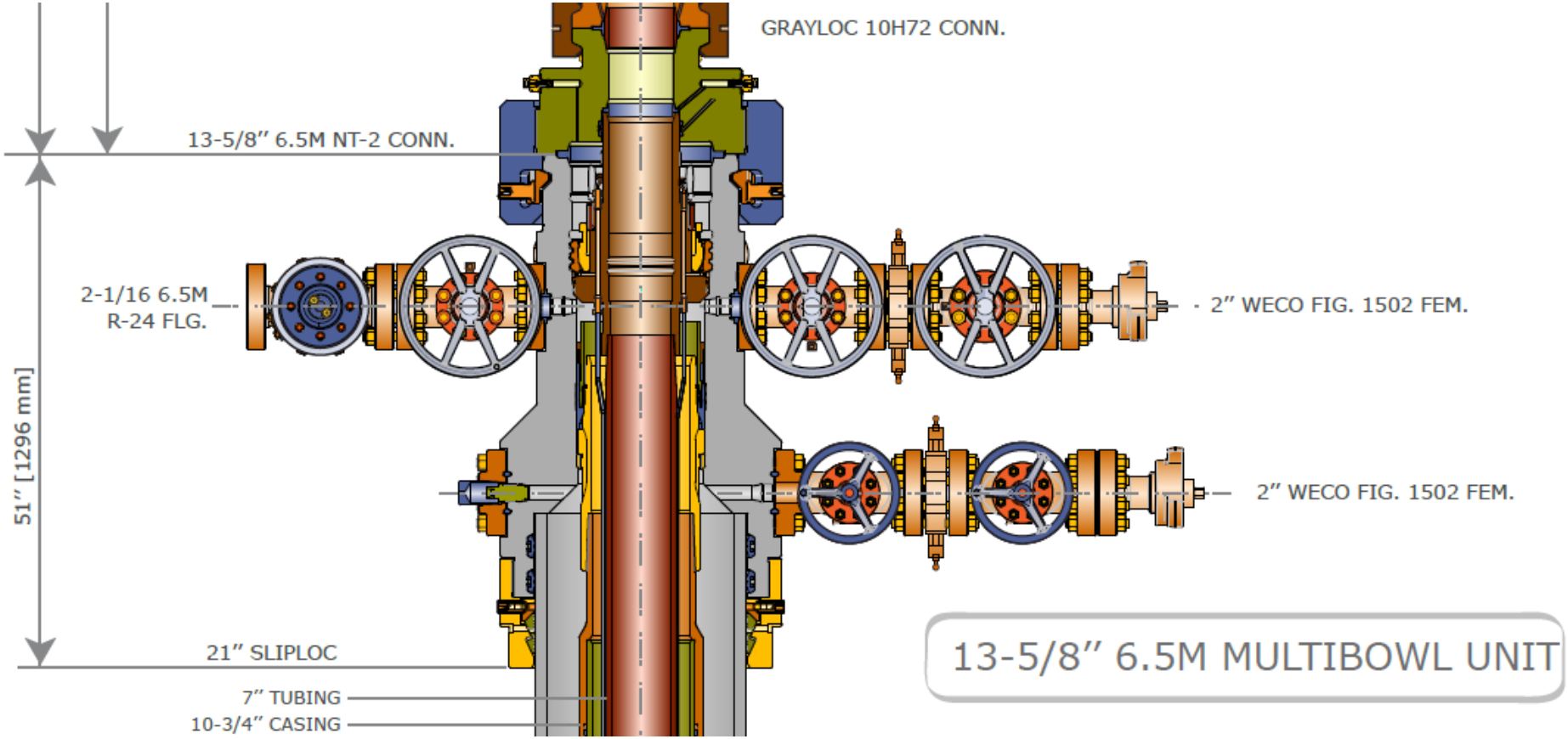
Asser	Symbol	Symbol Extra Info	MD [RKB] Top [m]	Length [m]	ID [inch]	Drift ID [inch]	Max OD [inch]	Description	MD [RKB] Bottom [m]	MD [Skid deck] [m]	TVD [RKB] [m]	Nom OD [inch]	Nom. weight [lb/ft]	Comment
			20.552	1.460	6.084	6.024	13.500	7" Tubing hanger Vetco Gray Volve	22.012	20.552	20.552	13.500	32.00	
			22.012	6.093	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	28.105	22.012	22.012	7.000	29.00	
			28.105	0.984	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	29.089	28.105	28.105	7.000	29.00	
		Cement: 20.510 - 202.000	29.089	471.122	6.184	6.059	7.644	7" 29# Vam Top Tubing	500.211	29.089	29.089	7.000	29.00	
			500.211	3.058	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	503.269	500.211	500.118	7.000	29.00	
		Control Line: 3/8" Hydraulic	503.269	2.775	5.812	5.810	9.500	Baker Mod: ONYX-7.5" TRSV Sz.: 5.812"	506.044	503.269	503.174	7.000	29.00	
			506.044	2.053	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	508.097	506.044	505.947	7.000	29.00	
		Casing: 10.75 20.510 - 600.000 Cement: 20.510 - 1363.000	508.097	2522.052	6.184	6.059	7.644	7" 29# Vam Top Tubing	3030.149	508.097	507.998	7.000	29.00	
		Cement: 2268.000 - 2588.000	3030.149	3.050	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	3033.199	3030.149	2704.828	7.000	29.00	
			3033.199	2.200	5.900	5.869	8.311	7" Single Gauge Mandrel for single NPQG-BA Tubing P/T Gauge	3035.399	3033.199	2706.266	7.000	32.00	
		Control Line: 1/4" Electrical cable	3035.399	2.050	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	3037.449	3035.399	2707.303	7.000	29.00	
			3037.449	23.209	6.184	6.059	7.644	7" 29# Vam Top Tubing	3060.658	3037.449	2708.270	7.000	29.00	
			3060.658	3.010	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	3063.668	3060.658	2719.207	7.000	29.00	
			3063.668	0.334	6.094	5.969	7.700	7" 29# Vam Top Collar	3064.002	3063.668	2720.623	7.000	32.00	
			3064.002	1.180	6.000	5.969	8.310	Granite Packer 9 5/8" x 7" (Size 831 x 600)	3065.182	3064.002	2720.780	7.000		
			3065.182	2.008	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	3067.190	3065.182	2721.335	7.000	29.00	
			3067.190	11.909	6.184	6.059	7.644	7" 29# Vam Top Tubing	3079.099	3067.190	2722.279	7.000	29.00	
			3079.099	12.000	6.184	6.059	7.680	7" 29# Vam Top Pup Joint B x P	3091.099	3079.099	2727.869	7.000	29.00	
			3091.099	2.054	6.184	6.089	7.690	7" 29# Vam Top Perforated Joint	3093.153	3091.099	2733.492	7.000	29.00	
			3093.153	0.295	4.882	4.767	6.190	7" 29# Locator Sub	3093.448	3093.153	2734.454	7.000	29.00	
			3093.448	4.300	6.184	6.059	7.050	7" 29# 13 CR Muleshoe	3097.748	3093.448	2734.592	7.000	29.00	

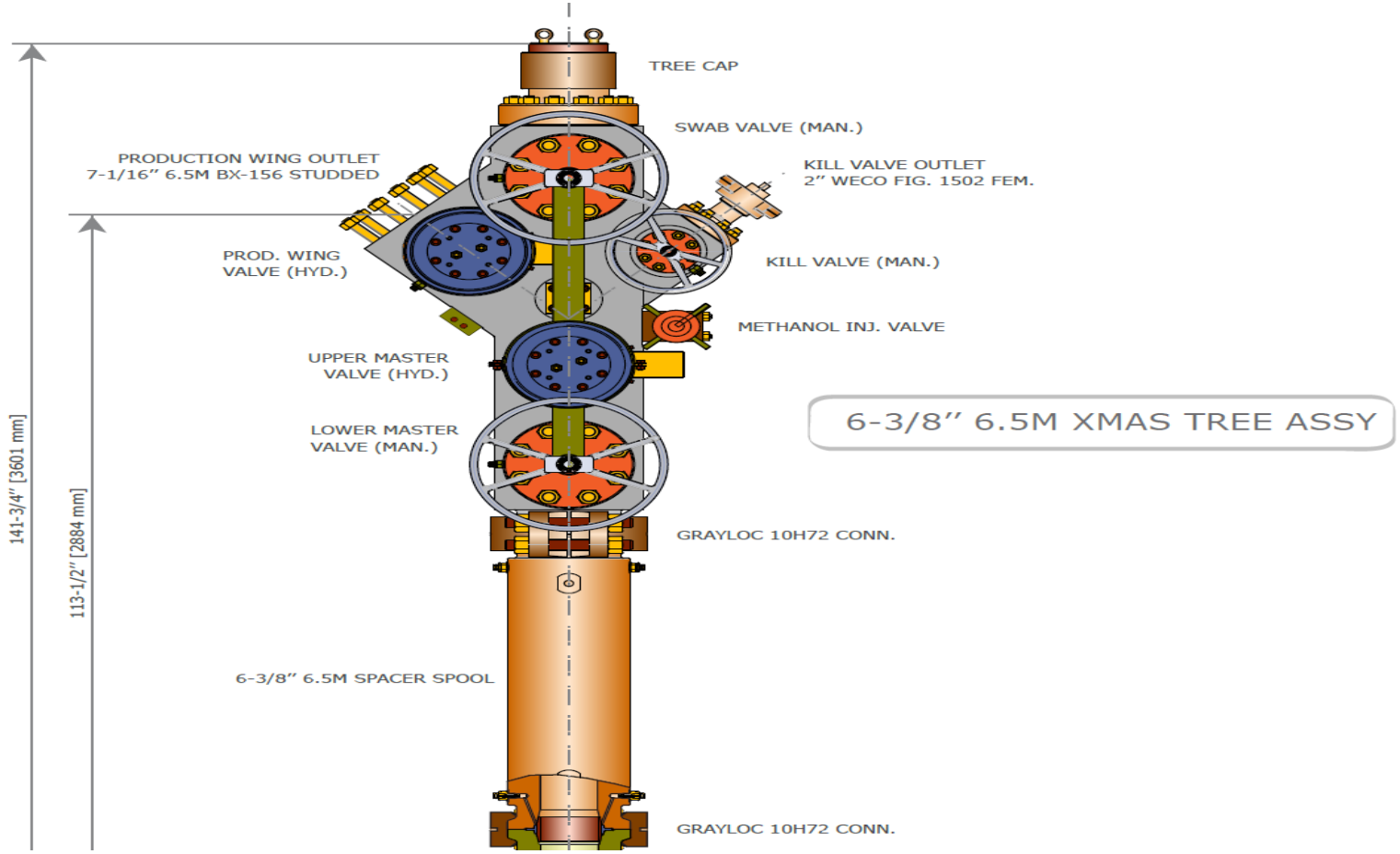
8.11 Wellhead / TSR / X-tree schematic



MAJOR ITEMS LIST

ITEM No.	DESCRIPTION	PART No.
1	30" MS-700 WELLHEAD HOUSING	N102810-1
2	18-3/4" MS-700 WELLHEAD HOUSING	N102880-1
3	13-3/8" MS-700 CASING HANGER	H57106-5
4	18-3/4" MS-1 ANNULUS SEAL	H57028-1
5	10-3/4" MS-700 TIEBACK CASING HANGER	N102818-1
6	10-3/4" TIEBACK ADAPTER	A50136-39
7	CASING HANGER LOCKDOWN SUB	H120090-1


GRAYLOC 10H72 CONN.
13-5/8" 6.5M NT-2 CONN.
2-1/16 6.5M
R-24 FLG.
2" WECO FIG. 1502 FEM.
2" WECO FIG. 1502 FEM.
51" [1296 mm]
21" SLIPLOC
7" TUBING
10-3/4" CASING
13-5/8" 6.5M MULTIBOWL UNIT


TREE CAP
SWAB VALVE (MAN.)
PRODUCTION WING OUTLET
7-1/16" 6.5M BX-156 STUDDED
KILL VALVE OUTLET
2" WECO FIG. 1502 FEM.
PROD. WING
VALVE (HYD.)
KILL VALVE (MAN.)
METHANOL INJ. VALVE
UPPER MASTER
VALVE (HYD.)
LOWER MASTER
VALVE (MAN.)
6-3/8" 6.5M XMAS TREE ASSY
141-3/4" [3601 mm]
113-1/2" [2884 mm]
GRAYLOC 10H72 CONN.
6-3/8" 6.5M SPACER SPOOL
GRAYLOC 10H72 CONN.

9 Appendix 4: Contractors list (optional)

Service	Contract no	Contractor
Drilling Fluids	SAP4600007720	Halliburton
Electric Wireline Logging	SAP4600014591	Schlumberger
Mud Logging	SAP4600017565	Baker Hughes Norge AS
Directional Drilling	SAP4600017565	Baker Hughes Norge AS
Wireline	SAP460001246x	Aker Well Service
Measurement While Drilling	SAP4600017565	Baker Hughes Norge AS
Rig Operations	SAP4500847188	Maersk Drilling
Tractor	SAP4600017779	Aker Well Service
Surf. WH/X-mas tree	SAP4600003983	Vetco Gray
Casing	SAP4600011449	Weatherford Norge AS

10 Appendix 5: Distribution list (optional)

No	Recipient	Number of copies